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Clinical Technique for Direct Composite Resin and Glass Ionomer Restorations

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The search for an ideal esthetic material for restoring teeth has resulted in significant improvements in resin-based composite materials.[1-4] Although these materials are referred to as resin-based composites, composite resins, and other terms, this book refers to most direct esthetic restorations as composites. Chapter 13 presents information about composite development, types, classification, material properties, and so forth. This chapter presents many of the specific clinical applications of these materials in both anterior and posterior teeth (Fig. 8.1). This chapter also presents the clinical technique for glass ionomer restorations.

The choice of a material to restore caries lesions and other defects in teeth is not always simple. Tooth-colored materials, such as composite, are used in almost all types and sizes of restorations. Such restorations are accomplished with minimal loss of tooth structure, little or no discomfort, relatively short operating time, and modest expense to the patient compared with indirect restorations. When a tooth is significantly weakened by extensive defects, however, and esthetics is of primary concern, the best treatment may involve the use of a ceramic onlay or crown.

The lifespan of a direct esthetic restoration depends on many factors, including the nature and extent of the initial caries lesion or defect; the treatment procedure; the restorative material and technique used; operator skill; and patient factors such as oral hygiene, occlusion, caries risk, and adverse habits.[5-9] Because all direct esthetic restorations are bonded to tooth structure, the effectiveness of generating the bond is paramount for the success and longevity of such restorations. Failures can result from numerous causes, including trauma, improper tooth preparation, inferior materials, poor material selection, and patient-related risk factors.

Composites can be used in almost any tooth surface. Naturally, factors must be considered for each specific application. The reasons for such expanded usage of these materials relate to improvements in their ability to bond to tooth structure (enamel and dentin) and in their physical and mechanical properties. The ability to bond a relatively strong material to tooth structure results in a restored tooth that is well sealed and regains a portion of its strength.[10,11]

General Considerations for Composite Restorations

This section summarizes general considerations about all composite restorations. A successful composite restoration requires careful attention to technique detail, so as to gain the maximum benefit of the material's properties. In selecting a direct restorative material, clinicians usually choose between composite and amalgam. Consequently, some of the following information provides comparative analyses between those two materials.

Indications and Contraindications

The *indications* for direct composites are:

1. Class I, II, III, IV, V, and VI restorations
2. Foundations and core buildups
3. Sealants and preventive resin restorations (conservative composite restorations)
4. Esthetic enhancement procedures:
   Partial veneers
   Full veneers
   Tooth contour modifications
   Diastema closures
5. Temporary or provisional restorations
6. Periodontal splinting
7. Luting of indirect esthetic restorations (when used in flowable form, or when heated to increase flow)

The primary *contraindications* for use of direct composites relate to:

1. Inability to obtain adequate isolation
2. Occlusal considerations related to wear and fracture of the composite material
3. Extension of the restoration on root surface
4. Operator factors

If the operating site cannot be isolated from contamination by oral fluids, composite (or any other bonded material) should not be used. If all of the occlusion is on the restorative material, composite may not be the choice for use particularly in patients with heavy occlusal function; however, the need to strengthen remaining weakened unprepared tooth structure with an economical composite restoration procedure (compared with an indirect restoration) and the commitment to recall the patient routinely and in a timely manner may override any concern about excessive wear potential. Any restoration that extends onto the root surface may result in less than ideal marginal integrity. Lastly, the operator must be committed to pursuing procedures, such as tooth isolation, that make bonded restorations successful. These additional procedures may make successful



• **Fig. 8.1** Composite restorations. A and B, Class II composite restoration, before and after. C and D, Class IV composite restoration, before and after.

bonded restorations more difficult and time consuming to achieve.

Advantages and Disadvantages

The following are *advantages* of composite restorations:

1. Esthetics
2. Conservative tooth preparation (less extension, minimum depth not necessary, mechanical retention usually not necessary)
3. Low thermal conductivity
4. Universal use
5. Adhesion to the tooth
6. Repairability

The primary *disadvantages* of composite restorations relate to their dependence on adequate adhesion and polymerization protocols and procedural difficulties. Composite restorations:

1. May have poor marginal and internal cavity adaptation, usually occurring on root surfaces as a result of polymerization shrinkage stresses or improper insertion of the composite
2. May exhibit marginal deterioration over time in areas where no marginal enamel is available for bonding
3. Are more difficult and time consuming to place, and more costly (compared with amalgam restorations) because bonding usually requires multiple steps; insertion is more difficult; establishing proximal contacts, axial contours, embrasures, and occlusal contacts may be more difficult; and finishing and polishing procedures are more difficult
4. Are more technique sensitive because the operating site must be appropriately isolated, incremental placement technique must be used for most materials, and proper adhesive technique is absolutely mandatory
5. May exhibit greater occlusal wear in areas of high occlusal stress or when all of the tooth's occlusal contacts are on the composite material

Clinical Technique for Pit-and-Fissure Sealants

Pits and fissures typically result from an incomplete coalescence of enamel and are particularly prone to caries. These areas can be sealed with a low-viscosity fluid resin after acid etching. Long-term clinical studies indicate that pit-and-fissure sealants provide a safe and effective method of preventing caries.[12-14] In children, sealants are most effective when applied to the pits and fissures of permanent posterior teeth immediately on eruption of the clinical crowns, provided proper isolation can be achieved. Adults also can benefit from the use of sealants if the individual experiences an increase in caries susceptibility because of a change in diet, lack of adequate saliva, or a particular medical condition.

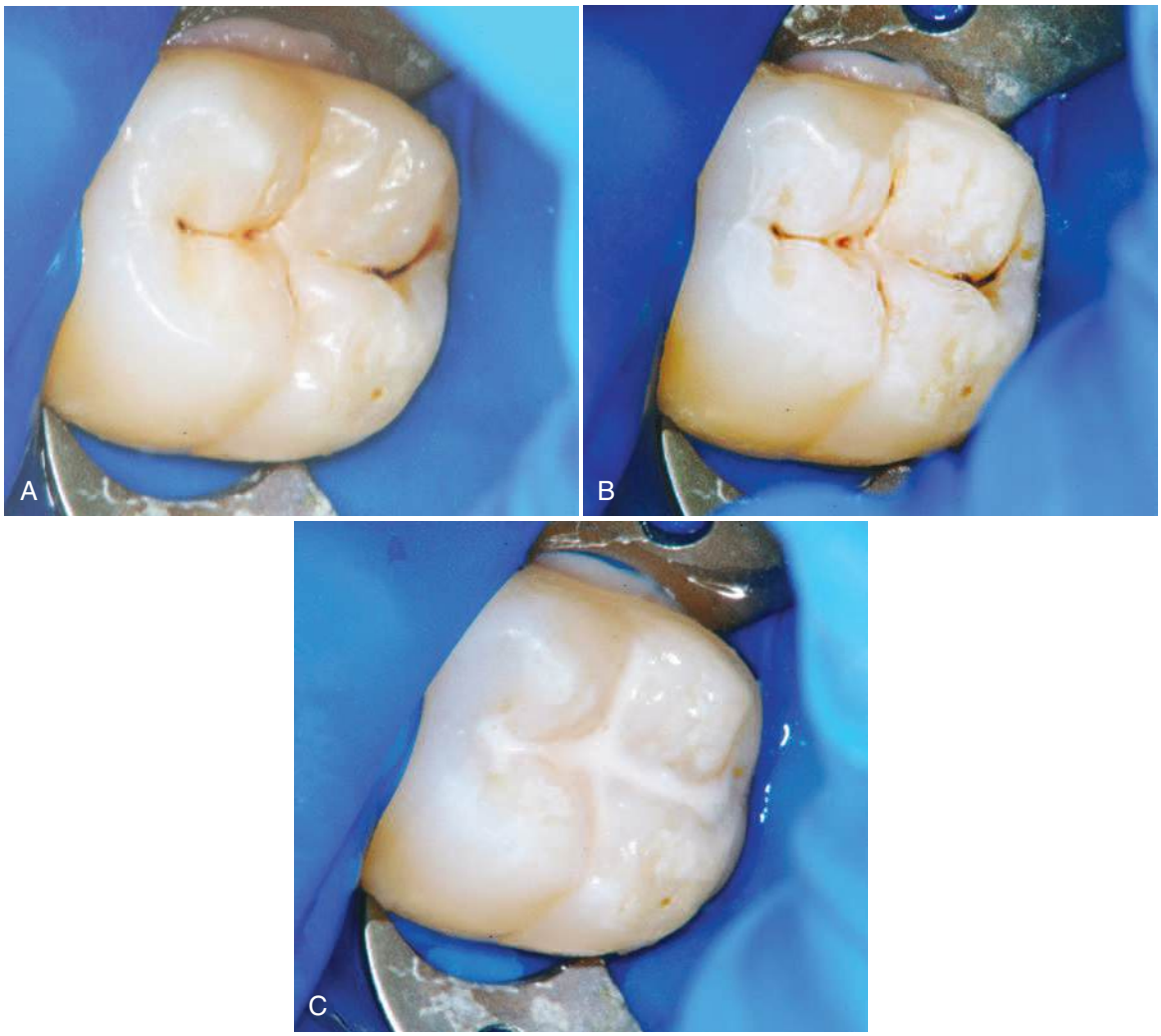
Sealants are indicated, regardless of the patient's age, for either preventive or therapeutic uses, depending on the patient's caries risk, tooth morphology, or presence of incipient enamel caries. In assessing the occlusal surfaces of posterior teeth as potential candidates for a sealant procedure, the primary decision is based on

whether a cavitated lesion exists. This decision is based primarily on radiographic and clinical examinations, although other adjunctive technologies for occlusal caries detection are available. Explorers must be used judiciously in the detection of caries, as a sharp explorer tine may cause a cavitation. The clinical examination should be primarily focused on visual assessments of a clean, dry tooth surface preferably under adequate light and magnification. The patient's caries risk also should be a factor when considering treatment options. See Chapter 2 for a discussion of emerging technologies for occlusal caries detection and monitoring.

When no cavitated caries lesion is diagnosed, the treatment decision is either to pursue no treatment or to place a pit-and-fissure sealant, particularly if the surface is at high risk for future caries. If a small caries lesion is detected, and the adjacent grooves and pits, although sound at the present time, are at risk for caries in the future, a preventive resin restoration or conservative composite restoration of the active cavitated lesion and placement of a sealant to include any radiating noncarious fissure or pits is recommended.

Although studies show that sealants can be applied over small, cavitated lesions, with no subsequent progression of the caries lesion, sealants should be used primarily for the prevention of caries rather than for the treatment of existing caries lesions.[15,16] A bitewing radiograph should be obtained and evaluated before sealant placement to ensure that no dentinal or proximal caries lesion is evident. Only caries-free pits and fissures or incipient lesions in enamel not extending to the dentinoenamel junction (DEJ) currently are recommended for treatment with pit-and-fissure sealants.[17]

Because materials and techniques vary, it is important to follow the manufacturer's instructions for the sealant material being used. A standard method for applying sealants to posterior teeth is presented here. Each quadrant is treated separately and one or more teeth may be sealed, as indicated. The following presentation relates to an occlusal surface of a mandibular first permanent molar (Fig. 8.2A). The tooth is isolated with a rubber dam (or another effective isolation method such as cotton rolls or Isolite). If proper isolation cannot be obtained, the bond of the sealant material to the tooth surface will be compromised, resulting in either loss of the sealant or caries under the inadequately bonded sealant. The area is cleaned with a slurry of pumice on a bristle brush and the tooth is rinsed thoroughly, while the explorer tip is used carefully to remove residual pumice or additional debris. The tooth surface is dried, and etched with 35% to 40% phosphoric acid for 15 to 30 seconds. Airborne particle abrasion techniques have been advocated for preparing pits and grooves before sealant placement, but their effectiveness has not been fully investigated.[18]



• **Fig. 8.2** Steps in application of pit-and-fissure sealant. A, After isolation and thorough cleaning of the occlusal surface to be sealed. B, After acid etching, rinsing, and drying. C, With sealant applied.

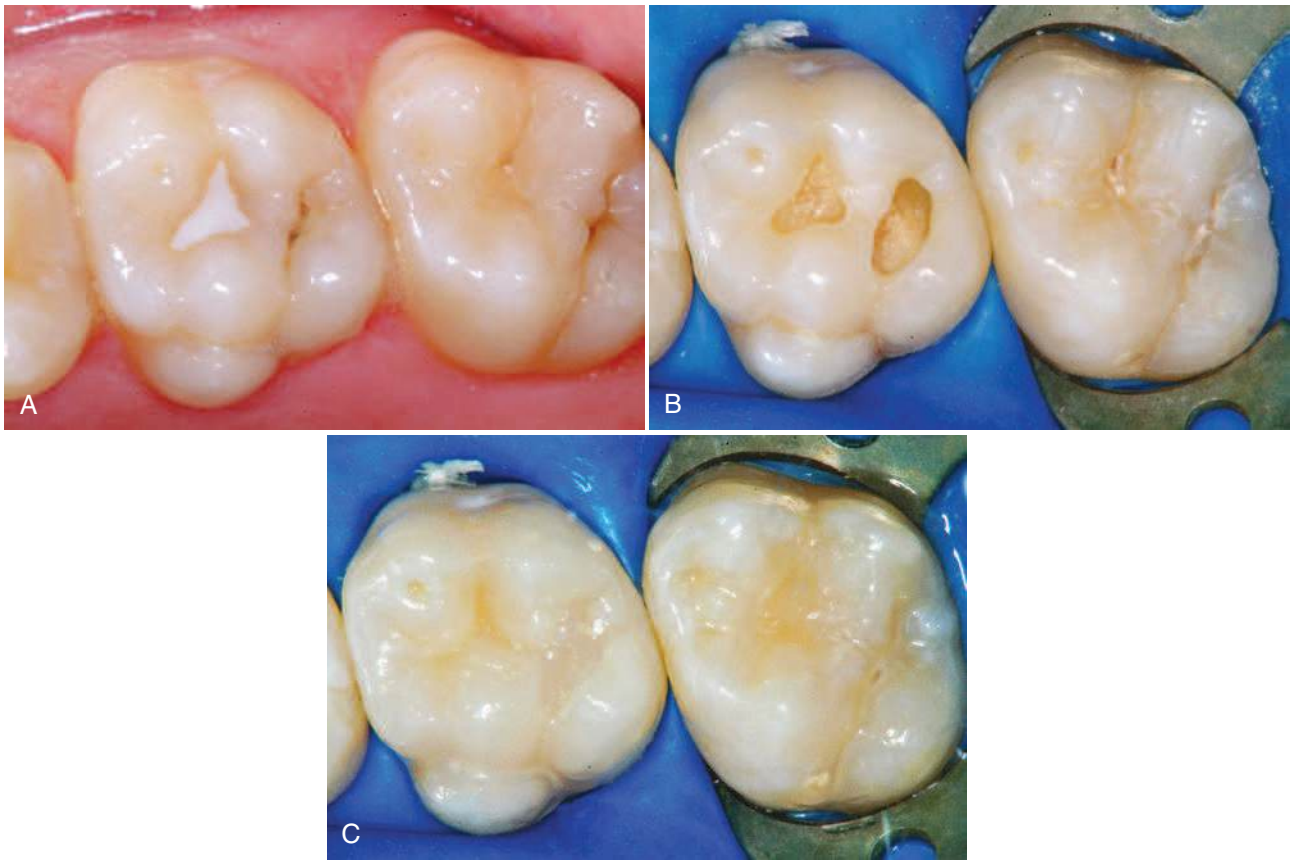
Properly acid-etched enamel surface has a lightly frosted (matte) appearance (see Fig. 8.2B). Any brown stains that originally may have been in the pits and fissures still may be present and should be allowed to remain. The sealant material is then applied and gently teased into place, to avoid entrapping air, and it should overfill slightly all pits and fissures, not extending onto unetched surfaces (see Fig. 8.2C). If too much sealant is applied, excess may be removed with a dry microbrush prior to light curing. After light curing and removal of the rubber dam, if used, the occlusion is evaluated by using articulating paper. If necessary, a round carbide finishing bur or white stone is used to remove any excess sealant. The surface usually does not require further polishing.

Clinical Technique for Preventive Resin and Conservative Composite Restorations

When restoring minimally carious pits and fissures on a previously unrestored tooth, an ultraconservative preparation design is recommended. This design allows for restoration of the caries lesion or defect with minimal removal of the tooth structure and often may be combined with the use of flowable composite or sealant to seal radiating noncarious fissures or pits that are at high risk for subsequent caries activity (Fig. 8.3). Originally referred to as a preventive resin restoration,[19-21] this type of ultraconservative restoration is now termed *conservative composite restoration* at the University of North Carolina.

An accurate diagnosis is essential before restoring the occlusal surface of a posterior tooth. The crucial factor in this clinical assessment is whether the suspicious pit or fissure has active caries that requires restorative intervention. Usually, a conservative composite restoration is the treatment of choice for the primary occlusal caries lesion as the tooth preparation can be minimally invasive.

If a definitive diagnosis of caries cannot be made for the entire occlusal surface, a conservative preparation of the suspicious area is performed with a small bur or diamond to determine the extent of the suspected fault without extending onto noncompromised fissures (Fig. 8.4). This approach is particularly indicated in patients with high caries activity and/or risk. As the tooth preparation is deepened, an assessment is made in the suspicious areas to determine whether to continue the preparation toward the DEJ (see Fig. 8.4C). If the suspicious fault is removed or found to be sound at a shallow preparation depth (minimal dentin caries), the conservative preparation and adjacent pits and fissures are etched with 35% to 40% phosphoric acid for 15 to 30 seconds, rinsed thoroughly, and lightly dried. The etched surfaces then are treated with an adhesive and restored with a flowable composite, which is placed and light cured according to manufacturer's instructions. The adjacent etched pits and fissures, if judged to be at risk, can be



• **Fig. 8.3** Conservative composite restoration. A, Occlusal view of the maxillary first and second molars. The first molar has caries on the distal occlusal pit and the second molar has suspicious pit on the disto-occlusal aspect. B, Caries was excavated from the first molar, and the second molar was minimally prepared. C, The first molar was restored with composite and the second molar received a conservative composite restoration with flowable composite.



• **Fig. 8.4** Class I direct composite restoration and conservative composite restoration. A, Mandibular second molar with suspicious occlusal pits; mandibular first molar with questionable sealant. B, After rubber dam isolation. C, Initial exploratory preparation reveals caries extending toward the dentinoenamel junction (DEJ). D, Conservative preparation on the second molar; the first molar was minimally prepared. E, Complete Class I direct composite restoration on the second molar; the first molar received a conservative composite restoration with flowable composite. F, Final restorations after the rubber dam was removed and the occlusion was checked.

sealed using a pit-and-fissure sealant or the same flowable composite following the technique described previously.

If the suspicious area is found to be carious, the preparation depth is extended until all of the soft carious dentin is removed (see Chapter 2 for a description of dentin caries zones), the prepared area is then restored with composite as described later in this chapter (Class I direct composite restoration), and unprepared pits and fissures are sealed. In the example presented in Fig. 8.4, the preparation was restored with composite.

Clinical Technique for Class I Direct Composite Restorations

Initial Clinical Procedures

A complete examination, diagnosis (including caries risk assessment), treatment plan, and informed consent should be finalized and obtained before the patient is scheduled for operative appointments (emergencies excepted). A brief review of the patient's record

(including medical factors), treatment plan, radiographs, and current caries risk should precede each restorative procedure.

Local Anesthesia

Local anesthesia is required for many operative procedures. Profound anesthesia contributes to a more comfortable and uninterrupted procedure and may result in reduction in salivation. These effects of local anesthesia contribute to better operative dentistry, especially when placing bonded restorations.

Preparation of the Operating Site

Prior to beginning any composite restoration, it may be necessary to clean the operating site with a slurry of pumice to remove plaque biofilm and superficial stains. Calculus removal with appropriate instruments also may be needed, although it would be preferable to see the patient after a thorough hygiene visit has been completed in this case. Prophy pastes containing flavoring agents, glycerin, or fluorides act as contaminants and must not be used as they will compromise the adhesive procedure.

Shade Selection

Although not as important for posterior when compared to anterior, more visible restorations, proper shade selection should be accomplished for all direct composite restorations. The shade of the tooth should be determined before teeth are subjected to any prolonged drying. Dehydrated teeth become lighter in shade as a result of a decrease in translucency secondary to water loss from the naturally porous tooth structure. A more comprehensive review of factors affecting shade selection is presented in Chapter 7, while esthetic considerations of tooth restoration are presented in Chapter 9.

Isolation of the Operating Site

Isolation for tooth-colored restorations is critical and can be accomplished with a rubber dam, an isolation device (e.g., Isolite), or cotton rolls/dry angles. Regardless of the method, isolation of the area is imperative if a successful bond is to be obtained. Contamination of etched enamel or dentin by saliva results in a significantly decreased bond; likewise, contamination of the composite material during insertion results in degradation of physical and mechanical properties.

Other Preoperative Considerations

When restoring posterior occlusal surfaces, a preoperative assessment of the occlusion should be made. This assessment should identify not only the occlusal contacts of the tooth or teeth to be restored but also the occlusal contacts on adjacent teeth. Knowing the preoperative location of occlusal contacts is important in planning the restoration outline form (so as to prevent an area of occlusal contact directly at a cavosurface/restoration interface) and establishing the proper occlusal contact on the restoration. Remembering where the contacts are located on adjacent teeth provides guidance in knowing when the restoration contacts are correctly adjusted.

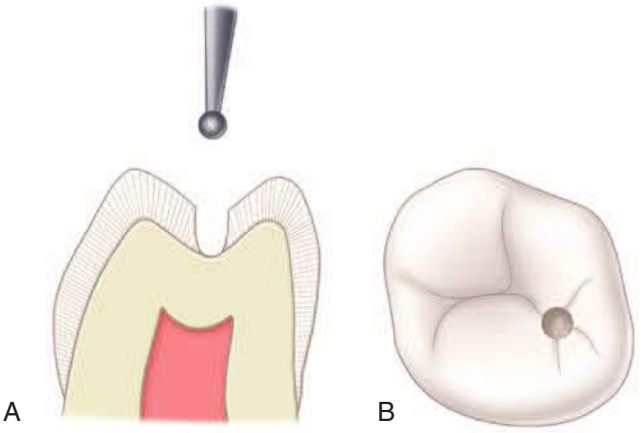
Tooth Preparation

As a general rule, the tooth preparation for direct posterior composites involves (1) creating access to the faulty structure, (2) removal of faulty structures (caries lesion, defective restoration, and base material, if present), and (3) creating convenience form for the restoration. When placing most posterior composites, it is not necessary to incorporate mechanical retention features in the tooth preparation.

Small to moderate Class I direct composite restorations may use minimally invasive tooth preparations and do not require typical resistance and retention form features. Instead, these conservative preparations typically result in more flared cavosurface forms without uniform or flat pulpal or axial walls. The initial pulpal depth is determined only by the selective removal of carious tooth structure, and there is no minimal thickness of restorative material requirement to limit bulk fracture.

These conservative preparations are prepared with a small round or elongated pear-shaped diamond or bur with round features, in an attempt to be as conservative as possible in the removal of the tooth structure. The size and shape of the instrument generally are dictated by the size of the lesion or other defect or by the type of defective restoration being replaced. If a round instrument is used, the resulting cavosurface margin angle may be more flared (obtuse) than if an elongated pear-shaped instrument is used (Fig. 8.5). Both carbide and diamond instruments can be used effectively. Although diamond instruments have been shown to create a thicker smear layer, thus raising concerns about the ability of self-etch adhesive systems to adequately reach and etch the underlying tooth structure, clinical trials have revealed excellent performance of mild self-etch adhesive systems suggesting this may not be a clinically relevant concern.[22-25]

Moderate to large Class I direct composite restorations, especially when used for larger caries lesions or to replace amalgam restorations, will typically feature flat walls that are perpendicular to occlusal forces, as well as strong tooth and restoration marginal configurations. All of these features help resist potential fracture in less conservative tooth preparations. However, the preparation should never be excessively extended beyond removal of faulty structures to justify resistance and retention forms, as this will further weaken the tooth structure and can ultimately lead to failure of the tooth-restoration unit. If the occlusal portion of the restoration is expected to be extensive, elongated pear-shaped cutting instruments with round features are preferred because they result in strong, 90-degree cavosurface margins without creating sharp internal line angles. However, this boxlike preparation form may increase the negative effects of the cavity configuration factor, or C-factor (see discussion in the next section). The objective of the tooth preparation is to remove all of the carious tissues or fault as conservatively as possible. Because the composite is bonded to the tooth structure, other less involved or at-risk areas can be sealed as part of the conservative

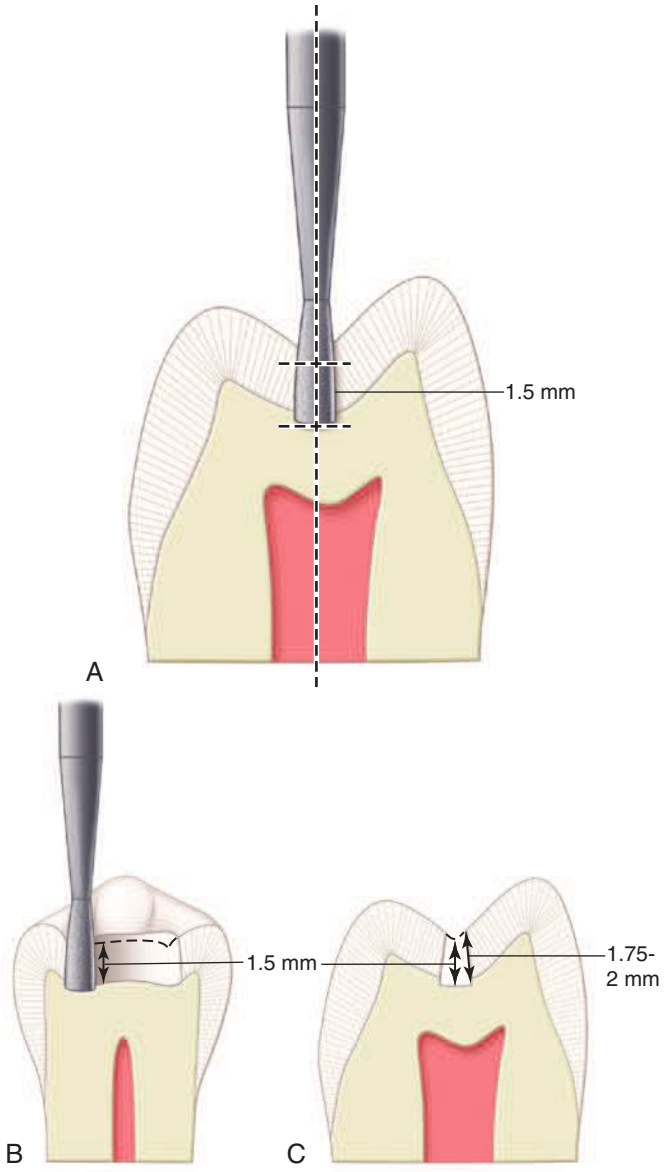


• **Fig. 8.5** Faciolingual cross section of small Class I tooth preparation using round diamond.

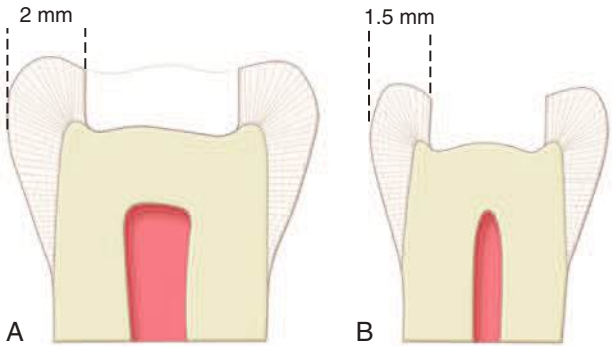
preparation techniques. Sealants may be combined with the Class I composite restoration, as described previously.

In moderate to large composite restorations, the tooth is entered in the area most affected by the caries lesion, with the elongated pear-shaped diamond or bur positioned parallel to the long axis of the tooth crown. When it is anticipated that the entire mesiodistal length of a central groove will be prepared, it is easier to enter the distal portion first and then transverse mesially to permit better operator visibility during the preparation. The pulpal floor is prepared to an initial depth that is approximately 0.2 mm internal to the carious DEJ (Fig. 8.6). Ideally the instrument is moved mesially, along the central groove in a controlled manner, to follow the rise and fall of the DEJ. The preparation is then carefully extended facially, lingually, mesially, and distally as indicated at the depth of the carious DEJ until a caries-free DEJ is identified around the whole periphery of the preparation. However, a flat pulpal floor with an initial preparation depth of 1.5 to 2 mm may also be acceptable (see Fig. 8.6B). Mesial, distal, facial, and lingual extensions are dictated by the caries lesion, old restorative material, or defect, always using the DEJ as a reference for both extensions and pulpal depth. Unnecessary extension into cuspal and marginal ridge areas should be avoided as much as possible as this compromises the strength of the tooth. Although the final bonded composite restoration may help restore some of the strength of weakened, unprepared tooth structure, the outline form should be as conservative as possible. Extensions into marginal ridges should result in at least 1.5 mm of remaining tooth structure (measured from the internal extension to the proximal height of contour) for premolars and approximately 2 mm for molars (Fig. 8.7). These limited extensions help preserve the dentinal support of the marginal ridge enamel and cusp tips and thus the overall ability of the tooth to resist occlusal forces.

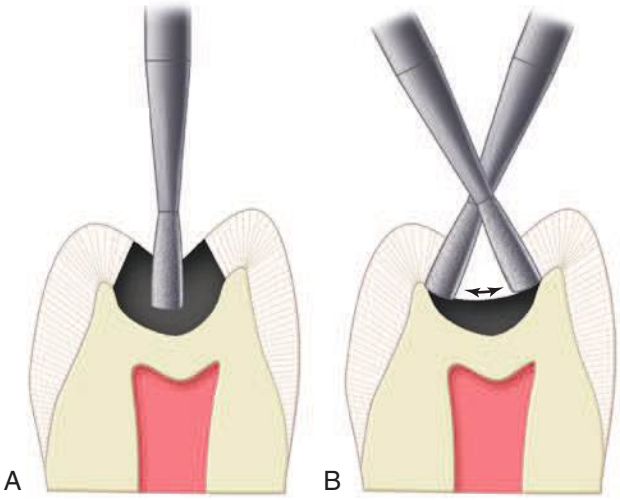
As the instrument is moved along the central groove, the resulting pulpal floor is usually moderately flat and follows the rise and fall of the DEJ (Fig. 8.8A). If extension is required toward the cusp tips, the same depth that is approximately 0.2 mm inside the DEJ



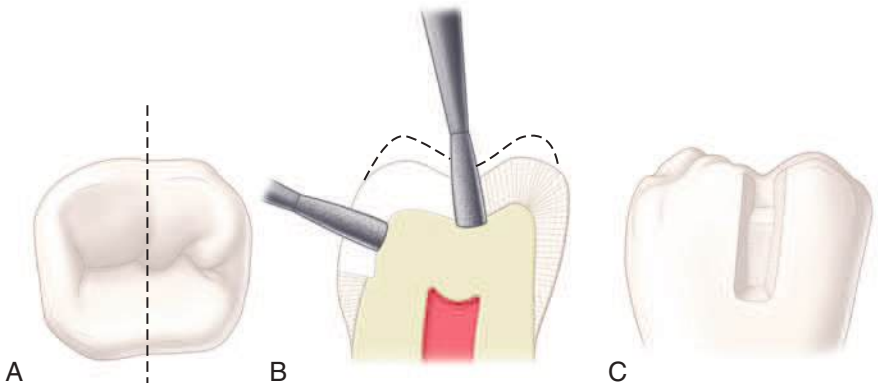
• **Fig. 8.6** A, Entry cut. Diamond or bur held parallel to the long axis of the crown. Initial pulpal depth is 1.5 mm from the central groove. When the central groove is removed, facial and lingual wall measurements usually are greater than 1.5 mm. (The steeper the wall, the greater is the height.) B, 1.5-mm depth from the central groove. C, Approximately 1.75- to 2-mm facial or lingual wall heights.



• **Fig. 8.7** Mesiodistal extension. Preserve dentin support of marginal ridge enamel. A, Molar. B, Premolar.



• **Fig. 8.8** A, After initial entry cut at correct initial depth (1.5 mm), caries remains facially and lingually. B, Orientation of diamond or bur must be tilted as the instrument is extended facially or lingually to maintain a 1.5-mm depth.



• **Fig. 8.9** Groove extension. A, Cross section through the faciolingual groove area. B, Extension through cusp ridge at 1.5-mm initial pulpal depth; the facial wall depth is 0.2 mm inside the dentinoenamel junction (DEJ). C, Facial view.

is maintained, usually resulting in the pulpal floor rising occlusally (see Fig. 8.8B). The same uniform depth concept also is appropriate when extending a facial or lingual groove radiating from the occlusal surface. When a groove extension is through the cusp ridge, the instrument prepares the facial (or lingual) portion of the faulty groove at an axial depth of 0.2 mm inside the DEJ and gingivally to include all caries and other defects (Fig. 8.9).

After extending the outline form to sound tooth structure, if any caries or old restorative material remains on the pulpal floor, it should be removed with the appropriately sized round bur or hand instrument. The occlusal margin is left as prepared. No attempt is made to bevel the occlusal margin because it may result in enlarging the occlusal isthmus of the preparation. Because of the occlusal surface enamel rod direction, the ends of the enamel rods already are exposed by the preparation, which further reduces the need for occlusal bevels.

Although large, extensive posterior composite restorations may have some potential disadvantages when used routinely, "real world" dentistry sometimes necessitates esthetic treatment alternatives that may provide a needed service to the patient. Often, patients simply cannot afford a more expensive esthetic restoration, or they may have dental or medical conditions that preclude their placement. In such instances, large posterior composite restorations sometimes may be used as a reasonable alternative when more definitive indirect options are not possible or realistic.

Restorative Technique

Placement of the Adhesive

See Chapter 5 for a more extensive discussion on adhesives. The dental adhesive is applied to the entire preparation with a microbrush, in accordance with the manufacturer's instructions. After application, the adhesive is polymerized with a light-curing unit, as recommended by the manufacturer.

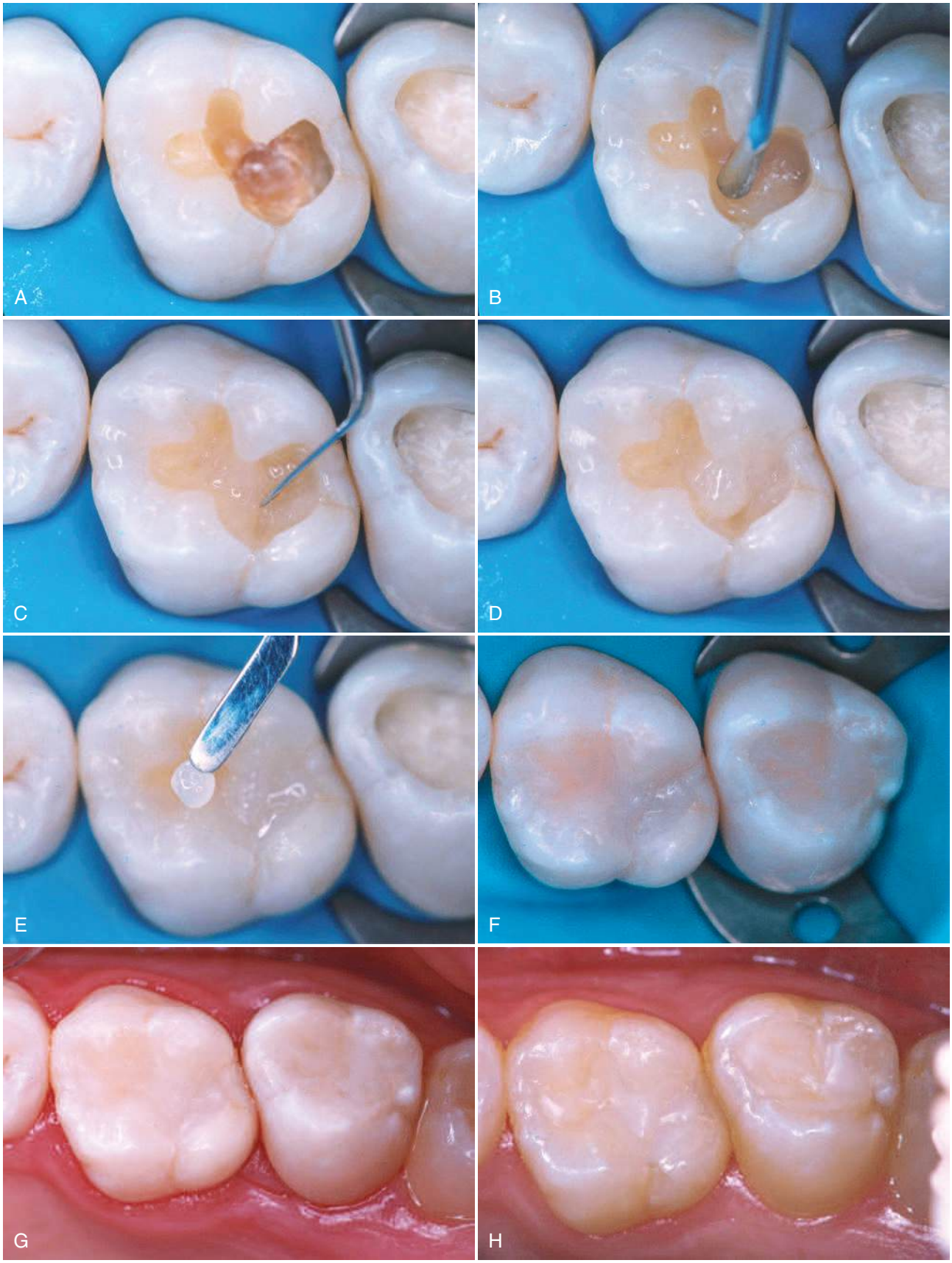
When the final tooth preparation is judged to be near the pulp in vital teeth, the operator may elect to use a base material prior to placing the adhesive and the composite. If the remaining dentin thickness (RDT) is clinically judged to be between 0.5 and 1.5 mm, a resin-modified glass ionomer (RMGI) base is used; if the RDT is judged to be less than 0.5 mm, a calcium hydroxide liner should be applied to the deepest aspect of the preparation, then protected with an RMGI base prior to adhesive placement.[26] In cases of mechanical pulp exposure, calcium hydroxide or, increasingly, mineral trioxide aggregate (MTA) can be used as a direct pulp-capping material.[26-28] If used, the calcium hydroxide or MTA liners should always be covered with a RMGI base, sealing the area and preventing the etchant (applied later) from dissolving the liner.[26,29]

Insertion and Light Curing of the Composite

A matrix is usually not necessary for Class I direct composite restorations, even when facial and lingual surface grooves are included. Composite insertion hand instruments or a compule may be used to place the composite material incrementally (Fig. 8.10). It is important to place (and light cure) the composite incrementally to ensure maximum polymerization and possibly to reduce the negative effects of polymerization shrinkage. Large increments or attempts at filling a preparation in bulk may compromise polymerization of the restoration secondary to inadequate depth of cure.

As mentioned, the term "C-factor" has been used to describe the ratio of bonded to unbonded surfaces in a tooth preparation and restoration. A typical Class I tooth preparation will have a high C-factor of 5/1 (i.e., five bonded surfaces—pulpal, facial, lingual, mesial, and distal—versus one unbonded surface—occlusal). The higher the C-factor of a tooth preparation, the higher the potential for composite polymerization shrinkage stress, as the composite shrinkage deformation is restricted by the bonded surfaces. The interface most likely to be adversely affected by high C-factor stresses is at the pulpal wall as the anatomy of dentin in this area makes establishment of a durable interface more difficult. Incremental insertion and light curing of the composite may reduce the negative C-factor effects for Class I composite restorations.[30-33]

The use of an RMGI liner or a flowable composite liner were thought to reduce the effects of polymerization shrinkage stress because of their favorable elastic modulus (a more elastic material would more effectively absorb polymerization stresses),[34,35] but those claims are not supported by current clinical evidence.[36] When composite is placed over an RMGI material, this technique is often referred to as a "sandwich" technique. The potential advantages of this technique are (1) the RMGI material bonds to the dentin without the need for a dental adhesive[37]; (2) the RMGI material, because of its bond to dentin and potential for fluoride release (potential anticariogenicity), may provide a better seal when used in cases where the preparation extends gingivally onto root structure[38]; and (3) the favorable elastic modulus of the RMGI may reduce the effects of polymerization shrinkage stresses. These suggested advantages are considered controversial, as no conclusive



• **Fig. 8.10** Class I composite incremental insertion. A, Tooth preparation for Class I direct composite restoration. B, After a resin-modified glass ionomer base is placed, the first composite increment is inserted and light cured. C–F, Composite is inserted and light cured incrementally, using cusp inclines as anatomic references to sculpt the composite before light curing. G, Completed restorations. H, At 5-year follow-up.



• **Fig. 8.11** Composite restorations. A and B, Class I composite, before and after. C and D, Class II composite, before and after.

published research based on longitudinal clinical trials evaluating the technique is available.

Regardless of the effect of incremental placement on shrinkage stress, posterior composites should be placed incrementally to facilitate proper light curing and development of correct anatomy. Especially in Class I direct composite restorations, the anatomic references of the occlusal unprepared tooth structure should guide the placement and shaping of the composite increments (Fig. 8.11; see Figs. 8.10G and 8.12I). If needed, very deep portions of the tooth preparation are restored first, with increments of no more than 2 mm in thickness (see Fig. 8.10B). The "enamel layer" of the restoration, that is, the occlusal increment(s), should be placed using an anatomic layering technique.[39] The operator places and shapes the composite before it is light cured so that the composite restores the occlusal anatomy of the tooth. Typically, the operator places and light cures one increment per cusp at a time and continues to place subsequent increments until the preparation is filled and the occlusal anatomy is fully developed (see Fig. 8.10C–F). The uncured composite can be shaped against the unprepared cusp inclines as visual guides, which will result in a very natural anatomic contour. Sculpture of the occlusal elements of the restoration following the anatomic references of the tooth respects the tooth's anatomy and occlusion and minimizes the need for contouring and finishing after the composite is polymerized. Furthermore, this technique prevents potential damage to the enamel adjacent to the restoration margins because it minimizes the need to use rotary instruments to remove excess composite extending beyond those margins. Any suitable composite hand instruments can be used with the anatomic layering technique. Fine composite spatulas and the tine of an explorer can be used to further develop or refine the anatomy of uncured composite increments. Microbrushes also can be used to smooth uncured composite against the preparation margin, but these should never be saturated with adhesives. Once the composite is fully cured, if additional contouring is needed, the restoration can be finished immediately after the last increment is cured using appropriate finishing instruments.

"Bulk-fill composites" have gained increased interest in recent years. These materials are available as *flowable base bulk-fill composites* and *full-body bulk-fill composites*.[40] Flowable base bulk-fill composites require a conventional composite as the occlusal increment and therefore are used for dentin replacement only, whereas full-body bulk-fill composites can replace dentin and enamel in a single increment. Their use can expedite the restorative procedure as increments of up to 4 mm in thickness are often suggested.[41] However, in vitro studies have raised concerns about a possible compromised internal adaptation and increased wear when these techniques are used.[42,43] Although this technique may present advantages, namely more expediency during composite placement, bulk-fill composites should be used with caution due to the lack of long-term clinical performance information at this time.[44] Bulk filling may also limit the operator's ability to carve the occlusal anatomy of the restoration before light curing.

Finishing and Polishing of the Composite

If the composite is carefully placed and shaped before light curing, as described in the previous section, additional finishing with burs is substantially minimized. However, in many cases, refined finishing

may be needed, especially when occlusion adjustments are necessary. The occlusal surface is shaped with a round or oval carbide finishing bur or similarly shaped finishing diamond. Polishing is accomplished with appropriate polishing cups, points, or both after the occlusion is adjusted as necessary (Fig. 8.12).

Clinical Technique for Class II Direct Composite Restorations

Initial Clinical Procedures

The same general procedures as described previously are necessary before beginning a Class II composite restoration. Several aspects of those activities, however, need emphasis. First, an assessment of the expected tooth preparation extensions (outline form) should be made and a decision rendered on whether or not an enamel periphery will exist on the tooth preparation, especially at the gingival margin. The expected presence of an enamel periphery strengthens the choice of composite as the restorative material because bonding to enamel is more predictable than bonding to dentin, especially along the gingival wall of the proximal preparation. If the preparation is expected to extend onto the root surface, potential problems with isolation of the operating area, adequate adhesion to the root dentin, and adequate composite polymerization exist. Good technique and proper use of the material may reduce these potential problems, but deep subgingival extension onto the root surface may be a contraindication for using composite resin as a posterior teeth restorative material.

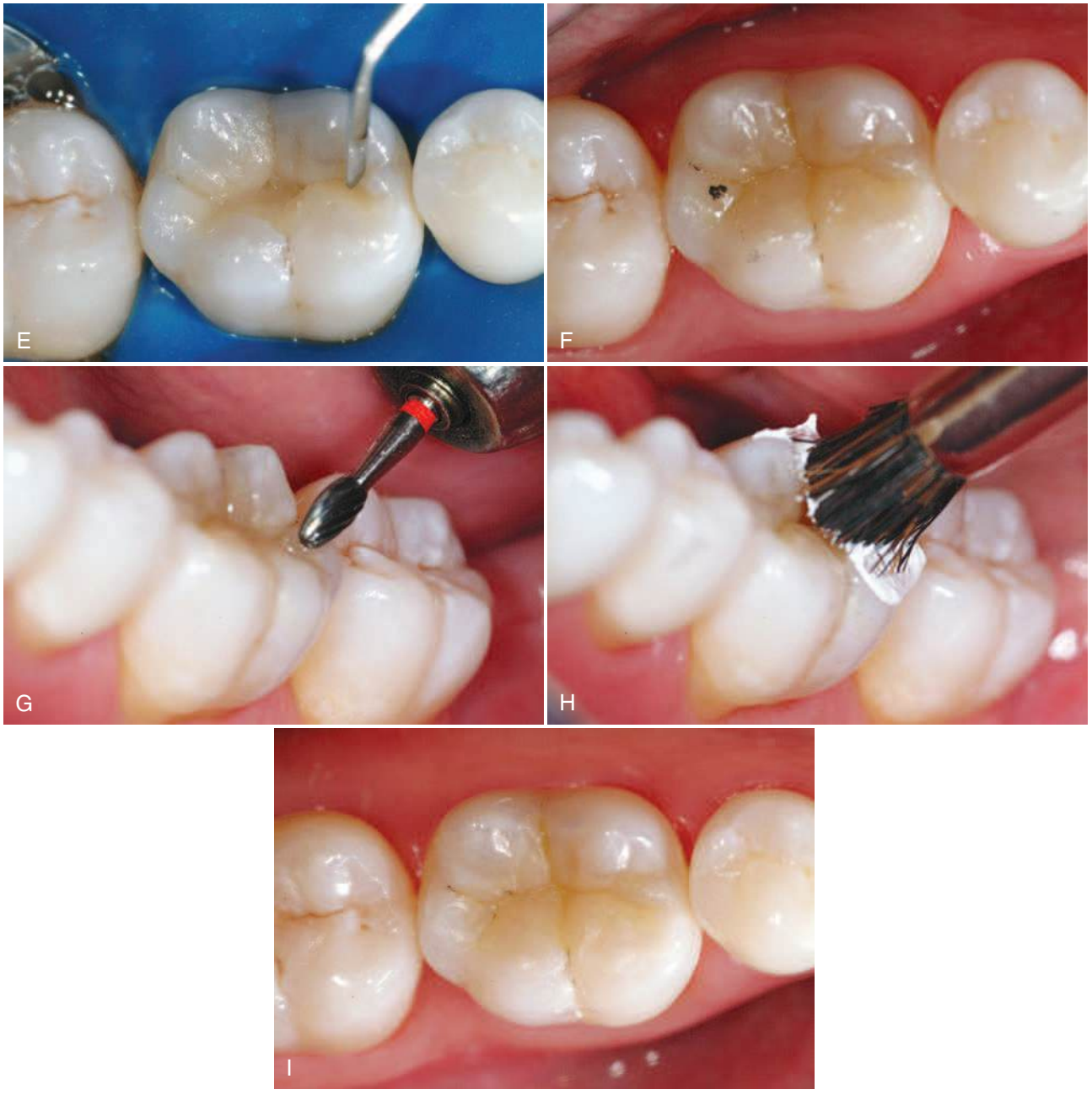
The preoperative occlusal relationship of the tooth to be restored must also be assessed. The presence of heavy occlusal contacts may indicate that wear may be more of a consideration. Also, preoperative wedging in the gingival embrasure of the proximal surfaces to be restored should occur. Placing wedges, bitine rings, or both before tooth preparation begins the separation of teeth, which may be beneficial in reestablishing the proximal contact with the composite restoration.

Tooth Preparation

Similar to the tooth preparation for Class I direct composite restorations, the tooth preparation for Class II direct composites involves (1) creating access to the faulty structure, (2) removal of faulty structures (the caries lesion, defective restoration, and base material, if present), and (3) creating the convenience form for the restoration. Retention, as with Class I restorations, is primarily obtained by bonding, so it is not necessary to use mechanical retention features in the tooth preparation for Class II composite restorations. Obtaining access to the defect may include removal of sound enamel to access the caries lesion. The extension of the preparation is therefore ultimately dictated by the extension of the fault or defect. It is usually not necessary to reduce sound tooth structure to provide "bulk for strength" or to provide conventional retention and resistance forms.



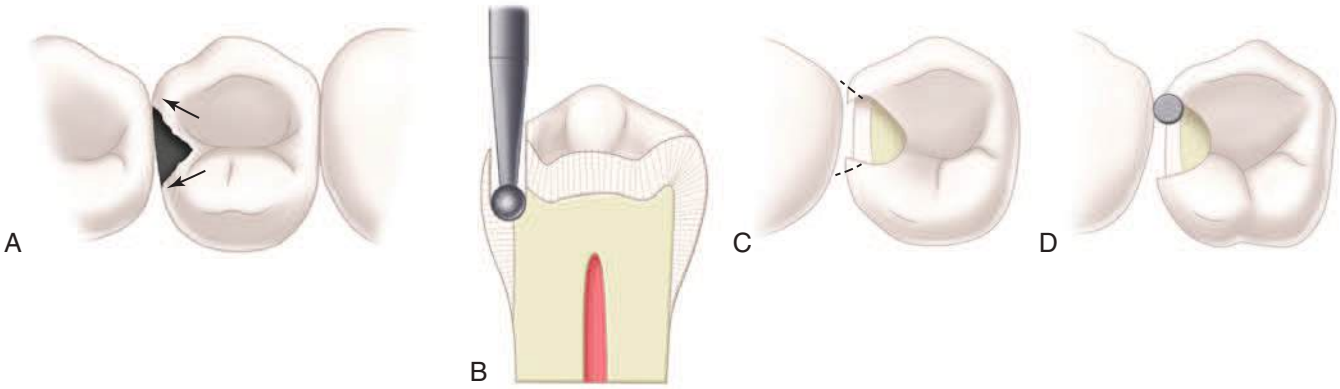
• **Fig. 8.12** Contouring and polishing of Class I composite. A, Mandibular molar with old amalgam restoration. B, Rubber dam isolation; old restoration is carefully removed to minimize increasing preparation size. C, Final tooth preparation. D, Incremental placement of composite. *Continued*



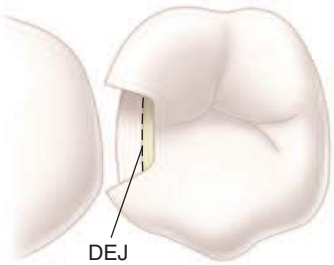
• **Fig. 8.12, cont'd** E, Incremental placement of composite. F, Rubber dam is removed and occlusion checked. G, Buccal view, a finishing fluted bur is used to selectively adjust the occlusion. H, Polishing with brush and diamond paste. I, Completed restoration.

Small Class II direct composite restorations are often used for primary caries lesions, that is, initial restorations. A small round or elongated pear-shaped diamond or bur with round features may be used for this preparation to remove the carious tissue or faulty material from the occlusal and proximal surfaces. To help prevent damage to the adjacent teeth and promote initial interproximal separation, wedges with or without stainless steel barriers may be utilized. The initial separation provided by wooden wedges will facilitate matrix placement in conservative preparations as well as result in tighter interproximal contacts. The pulpal and axial depths are dictated only by the depth of the lesion and are not necessarily uniform. The proximal extensions likewise are dictated only by the extent of the lesion but may require the use of another instrument with straight sides to prepare walls that are 90 degrees or greater (Fig. 8.13). The objective is to remove the caries lesion or defect conservatively, as well as any friable tooth structure.

Another conservative design for small Class II composites is the box-only tooth preparation (Fig. 8.14). This design is indicated when only the proximal surface is defective, with no lesions on the occlusal surface. A proximal box is prepared with a small, elongated pear-shaped or round instrument, held parallel to the long axis of the tooth crown. The instrument is extended through the marginal ridge in a gingival direction aiming at the center of the proximal caries lesion or defect. The axial depth is dictated by the extent of the caries lesion or fault. The form of the box depends on which instrument shape is used. The facial, lingual,



• **Fig. 8.13** Class II direct composite tooth preparation. A, Preoperative visualization of faciolingual proximal box extensions. Arrows indicate desired extensions. B, Round or oval, small elongated pearl instrument used. C and D, Facial, lingual, and gingival margins may need undermined cavosurface enamel (indicated by dotted lines) removed with straight-sided thin and flat-tipped rotary instrument or hand instrument.



• **Fig. 8.14** Box-only Class II composite preparation.

and gingival extensions are dictated by the extension of the caries lesion or defect. No beveling or secondary retention is indicated.

A third conservative design for restoring proximal lesions on posterior teeth is the facial or lingual slot preparation (Fig. 8.15). Here, a lesion is detected on the proximal surface, but the operator believes that access to the lesion can be obtained from either a facial direction or a lingual direction, rather than through the marginal ridge in a gingival direction. Usually, a small round diamond or bur is used to gain access to the lesion. The instrument is oriented at the correct occlusogingival position, and the entry is made with the instrument as close to the adjacent tooth as possible, preserving as much of the facial or lingual surface as possible. The preparation is extended occlusally, facially, and gingivally enough to remove the lesion. The axial depth is determined by the extent of the lesion. The occlusal, facial, and gingival cavosurface margins are 90 degrees or greater. Care should be taken not to undermine the marginal ridge during the preparation. If the defect extends occlusally to the point of undermining the marginal ridge, a more conventional Class II preparation, with an occlusal access component, should be used.

The tooth preparation for *moderate to large Class II direct composite restorations* has features that resemble a more traditional Class II amalgam tooth preparation and may include an occlusal step and a proximal box depending on the location, extension, and depth of the caries lesion.

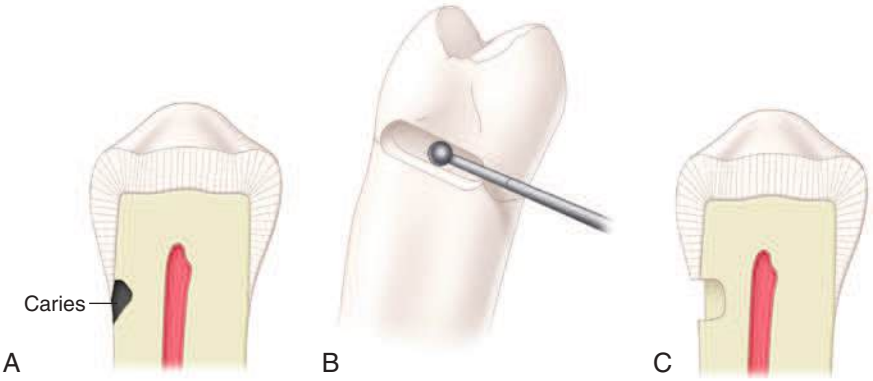
The occlusal portion of the Class II preparation is prepared similarly as described for the Class I preparation. The primary differences are related to technique of incorporating the faulty proximal surface. Preoperatively, the proposed facial and lingual proximal extensions should be visualized (see Fig. 8.13A). Initial occlusal extension toward the involved proximal surface should go through the marginal ridge area at initial pulpal floor depth, exposing the DEJ. The DEJ serves as a guide for preparing the proximal box portion of the preparation.

A No. 330 or No. 245 shaped diamond or bur is used to enter the pit next to the carious proximal surface. The instrument is positioned parallel with the long axis of the tooth crown. If only one proximal surface is being restored, the opposite marginal ridge dentinal support should be maintained (Fig. 8.16).

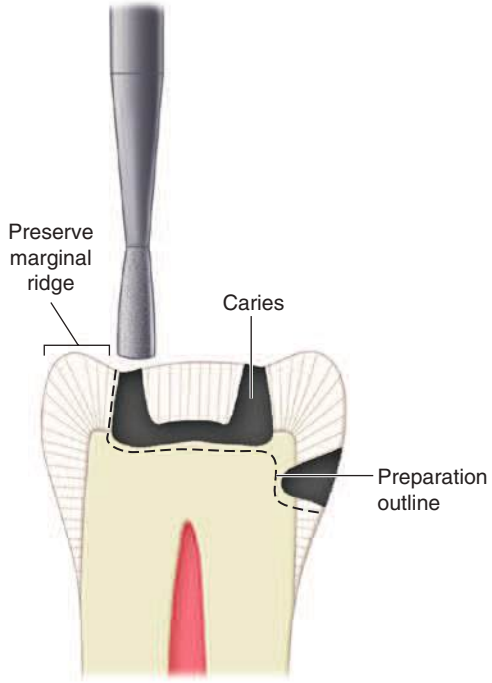
The pulpal floor is initially prepared with the instrument to a depth that is approximately 0.2 mm inside the DEJ. The instrument is moved to include the caries lesion and all defects facially or lingually, or both, as it transverses the central groove. Every effort should be made, however, to keep the faciolingual width of the preparation as narrow as possible. The initial depth is maintained during the mesiodistal movement, but follows the rise and fall of the underlying DEJ. The pulpal floor is relatively flat in a faciolingual plane but may rise and fall slightly in a mesiodistal plane (Fig. 8.17). If more of the caries lesion remains in dentin, it is removed after the preparation outline, including the proximal box extensions, has been established.

Because the facial and lingual proximal extensions of the faulty proximal surface were visualized preoperatively, the occlusal extension toward that proximal surface begins to widen facially and lingually to begin to outline those extensions as conservatively as possible. Care is taken to preserve cuspal areas as much as possible during these extensions. At the same time, the instrument extends through the marginal ridge to within 0.5 mm of the outer contour of the marginal ridge. This extension exposes the proximal DEJ and protects the adjacent tooth (see Fig. 8.17). At this time, the occlusal portion of the preparation is complete except for possible additional pulpal floor caries lesion excavation. The occlusal walls generally converge occlusally because of the inverted shape of the instrument.

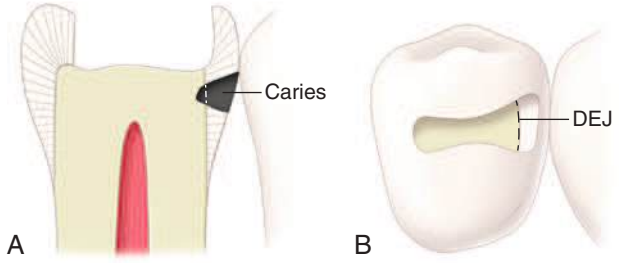
Typically, caries develops on a proximal surface immediately gingival to the proximal contact. The extent of the caries lesion and amount of old restorative material dictate the facial, lingual,



• **Fig. 8.15** Facial or lingual slot preparation. A, Cervical caries on the proximal surface. B, The round diamond or bur enters the tooth from the accessible embrasure, oriented to the occlusogingival middle of the lesion. C, Slot preparation.



• **Fig. 8.16** When only one proximal surface is affected, the opposite marginal ridge should be maintained.

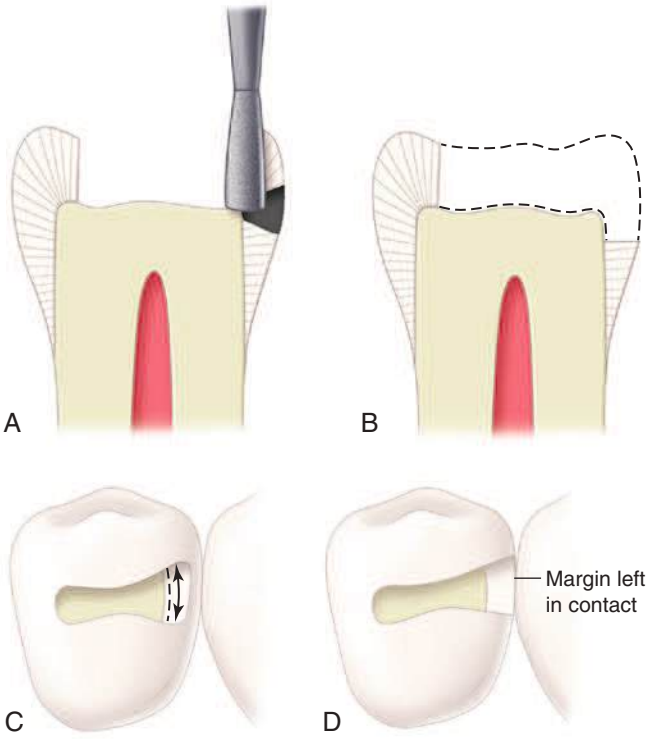


• **Fig. 8.17** Occlusal extension into faulty proximal surface. A and B, Extension exposes the dentinoenamel junction (DEJ) but does not hit the adjacent tooth. Facial and lingual extensions as preoperatively visualized (see Fig. 10.9 for initial pulpal floor depth).

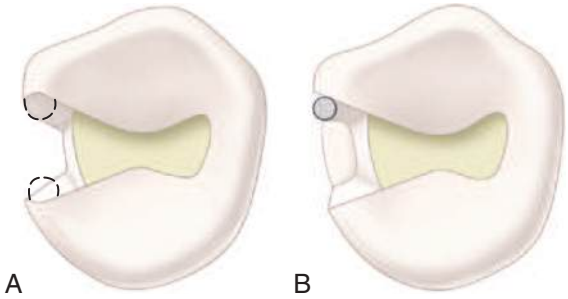
and gingival extensions of the proximal box of the preparation. Although it is not required to extend the proximal box beyond contact with the adjacent tooth (i.e., provide clearance with the adjacent tooth), it may simplify the preparation, matrix placement, and contouring procedures. If all of the defect can be removed without extending the proximal preparation beyond the contact, however, the restoration of the proximal contact with the composite is simplified (Fig. 8.18D).

Before the instrument is extended through the marginal ridge, the proximal ditch cut is initiated. The operator holds the instrument over the DEJ with the tip of the instrument positioned to create a gingivally directed cut that is 0.2 mm inside the DEJ (see Fig. 8.18B–D). For a No. 245 instrument with a tip diameter of 0.8 mm, this would require one fourth of the instrument's tip positioned over the dentin side of the DEJ (the other three fourths of the tip over the enamel side). The instrument is extended facially, lingually, and gingivally to include all of the caries lesion or old material, or both. The faciolingual cutting motion follows the DEJ and therefore is usually in a slightly convex arc outward (see Fig. 8.18C). During this entire cutting, the instrument is held parallel to the long axis of the tooth crown. The facial and lingual margins are extended as necessary and should result in at least a 90-degree margin (indicating enamel that is supported by dentin), more obtuse being acceptable as well. If the preparation is conservative, then a smaller, thinner instrument is used to complete the faciolingual wall formation, avoiding the creation of iatrogenic damage to the adjacent tooth (Fig. 8.19). Alternatively, a sharp hand instrument such as a chisel, hatchet, or a gingival margin trimmer can be used to finish the enamel wall. At this point, the remaining proximal enamel that was initially maintained to prevent damage to the adjacent tooth has been removed. The gingival floor is prepared flat with an approximately 90-degree cavosurface margin. Gingival extension should be as minimal as possible, to preserve marginal enamel. The axial wall should be 0.2 mm inside the DEJ and have a slight outward convexity. For large caries lesions, additional axial wall caries excavation may be necessary later (Fig. 8.20).

If no caries lesion in the dentin or other defect remains, the preparation is considered complete at this time (Fig. 8.21). However, if carious dentin or faulty restorative material/base remain on the axial and pulpal aspects of the preparation, which is typically the case for moderate to large Class II restorations, these areas should be selectively excavated to firm dentin. Often the initial preparation



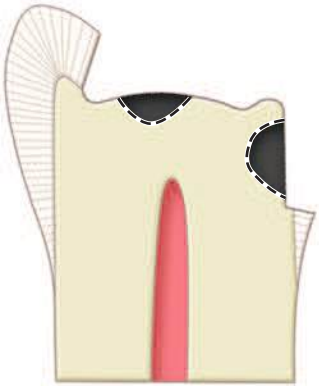
• **Fig. 8.18** A, The proximal wall may be left in contact with the adjacent tooth. B, Proximal ditch cut. The instrument is positioned such that a gingivally directed cut creates the axial wall 0.2 mm inside the dentinoenamel junction (DEJ). C, Faciolingual direction of axial wall preparation follows the DEJ. D, Axial wall 0.2 mm inside the DEJ.



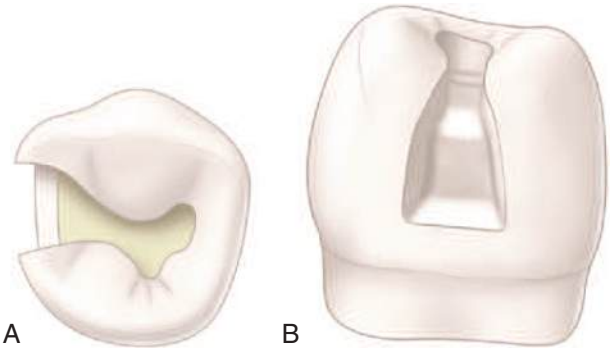
• **Fig. 8.19** Using a smaller instrument to prepare the cavosurface margin areas of facial and lingual proximal walls. A, Facial and lingual proximal margins undermined. B, Using a smaller instrument.

outline will have to be enlarged to allow access at this stage of the preparation. Once selective removal of any remaining carious tissue or removal of faulty existing restorative/base materials is accomplished, consideration should be given for pulp protection as described previously in the Class I restorative technique section of this chapter. When treating a tooth with very extensive and deep caries on a vital, asymptomatic tooth, care should be exercised to avoid a pulp exposure, which would considerably compromise the prognosis for the tooth. See Chapter 2 for considerations on selective removal of carious dentin.

Because the composite is retained in the preparation by adhesion, no secondary preparation retention features are necessary. No bevels are placed on the occlusal cavosurface margins because these walls already have exposed enamel rod ends. Beveled composite margins



• **Fig. 8.20** Proximal extension. The enamel margin on the gingival floor is critical for bonding, so it should be preserved, if not compromised. Any remaining soft dentin on the axial wall (or the pulpal floor) is excavated as part of the final tooth preparation (as indicated by dotted lines).



• **Fig. 8.21** Final Class II composite tooth preparation. A, Occlusal view. B, Proximal view.

also may be more difficult to finish. Bevels are rarely used on any of the proximal box walls because of the difficulty in restoring these areas, particularly when using inherently viscous composites. Bevels also are not recommended along the gingival margins of the proximal box; however, it is still necessary to remove brittle unsupported enamel rods along the margins because of the gingival orientation of the enamel rods. For most Class II preparations, this margin already is approaching the cementoenamel junction (CEJ), and the enamel is thin. Care is taken to maintain any enamel in this area to achieve a preparation with all-enamel margins. If the preparation extends onto the root surface, more attention must be focused on keeping the area isolated during the bonding technique, but no differences in tooth preparation are required. Usually, the only remaining final tooth preparation procedure that might be necessary is additional removal of the caries lesion associated with the dentin on either the pulpal floor or the axial wall. If necessary, a round bur or appropriate spoon excavator is used for selective removal of any remaining carious tissue.

Restorative Technique

Matrix Application

One of the most important steps in restoring Class II preparations with direct composites is proper matrix selection and setup. As with most restorative procedures involving a proximal surface, a matrix is necessary to (1) confine the restorative material excess and (2) assist in the development of the appropriate axial tooth

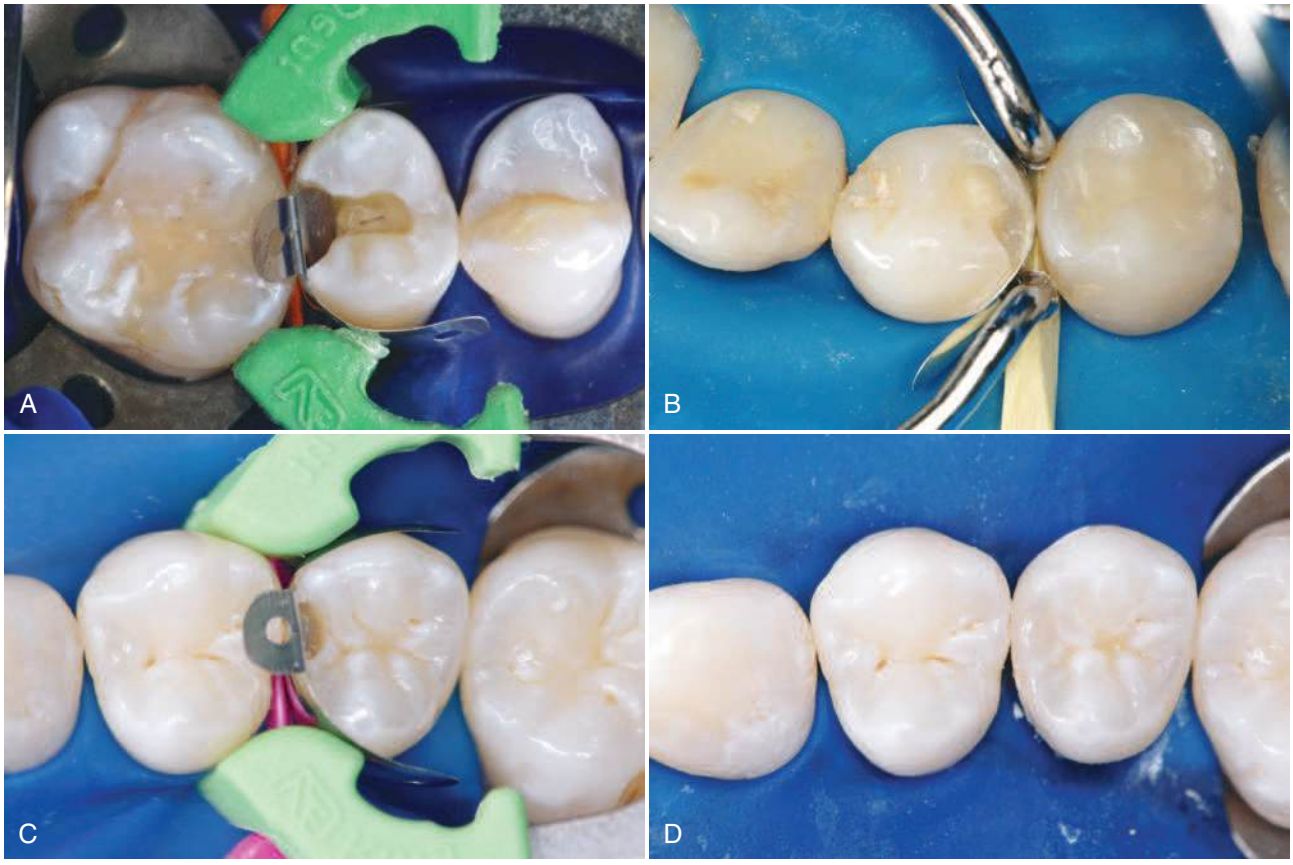
contours. In contrast to amalgam, which can be condensed against the matrix and thereby improve the proximal contact, Class II composites are almost totally dependent on the contour and position of the matrix for establishing appropriate proximal contacts. Early wedging and retightening of the wedge during tooth preparation aid in achieving sufficient separation of teeth to compensate for the thickness of the matrix material. Before placing the composite material, the matrix material must be in absolute contact with (i.e., touching) the adjacent contact area.

Generally, the matrix is applied before adhesive placement. An ultrathin metal matrix band generally is preferred for the restoration of a Class II composite because it is thinner than a typical metal band and can be contoured better than a clear polyester matrix. No significant problems are experienced in placing and light curing composite material when using a metal matrix as long as an incremental technique is used.

Although a Tofflemire-type matrix band can be used for restoring a two-surface tooth preparation, precontoured sectional metallic matrices are preferable (Fig. 8.22), because only one thickness of metal matrix material is encountered instead of two, making contact generation easier. These sectional matrices are relatively easy to use, very thin, and come in different sizes so as to accommodate for varied occlusogingival heights of the proximal box. There are several systems available, and selection is based on operator preference. These systems may use a bitine ring to (1) aid in stabilizing the sectional matrix and (2) provide additional tooth separation while the composite is inserted. The primary benefit of these systems is a simpler method for establishing an appropriate composite proximal contour and contact.[45] Use of these systems for restoring wide faciolingual proximal preparations requires careful application; otherwise the bitine ring prongs may cause deformation of the matrix band, resulting in poor restoration contour.

When both proximal surfaces are involved, a Tofflemire retainer with an ultrathin (0.025 mm or 0.001 inch) and burnishable matrix band is used. The band is contoured, positioned, wedged, and shaped, as needed, for proper proximal contacts and embrasures. Before placement, the metal matrix band for posterior composites should be burnished on a paper pad to impart proper proximal contour to the band. Alternatively, an ultrathin precontoured metal matrix band may be used in the Tofflemire retainer.

Regardless of the type of matrix system used, the matrix material should extend at least 1 mm beyond the gingival margin (gingivally) and the area corresponding to the marginal ridge of the restoration (occlusally). The matrix should not extend further subgingivally or occlusally so as not to interfere with the restorative procedure. If a wedge was used during the tooth preparation procedures, it is removed so that the matrix band can be installed, and immediately reinserted. A wedge is needed at the gingival margin to (1) hold



• **Fig. 8.22** Sectional matrix systems for posterior composites. A, Sectional matrix system in place with plastic wedge and bitine ring to restore the maxillary premolar with direct composite. B, Sectional matrix system in place with wooden wedge and bitine ring to restore the mandibular premolar with direct composite. C, Sectional matrix system in place with plastic wedge and bitine ring to restore the maxillary premolar with direct composite. D, Case presented in C after placement and light curing of the composite, and matrix removal, before any contouring. Note minimal excess composite as a result of good matrix adaptation to facial and lingual embrasures.

the matrix in position, (2) provide slight separation of the teeth, and (3) prevent a gingival overhang of the composite material. A wedge must be used to separate teeth sufficiently to compensate for the thickness of the matrix if the completed restoration is to have appropriate proximal contacts.

Several types of commercial wedges are available in assorted sizes. An anatomical (triangle-shaped) wedge of compatible size is indicated for most preparations. The wedge is kept as short as possible so as not to limit operator access to the preparation during insertion of the restorative material, and should also not push the matrix onto the preparation space.

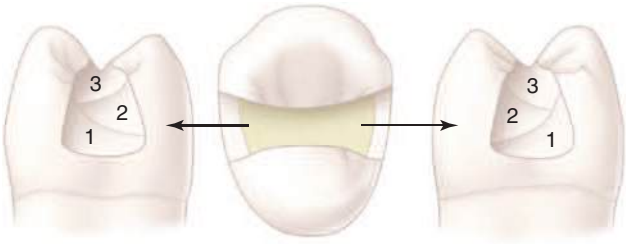
For posterior restorations, the wedge is placed from the larger to the smaller interproximal gingival embrasure, typically from a lingual approach, just apical to the gingival margin. The wedge should engage (stabilize) the matrix band. When a rubber dam is used, wedge placement may be aided by a small amount of water-soluble lubricant on the tip of the wedge. The rubber dam is first stretched gingivally (on the side from which the wedge is inserted), then released gradually during wedge insertion.

Placement of the Adhesive

The technique for adhesive placement is as described previously for the Class I direct composite restoration. Care should be exercised to avoid adhesive pooling along the matrix and preparation margins gingivally, facially, and lingually.

Insertion and Light Curing of the Composite

The technique for insertion of Class II posterior composites is illustrated in Fig. 8.23. It is best to restore the proximal box portion of the preparation first. Hand instruments or a "composite gun" may be used to insert the composite material. It is important to place and light cure the composite incrementally to maximize the curing potential and to reduce the negative effects of polymerization shrinkage as discussed earlier in this chapter. Many techniques have been described for the restoration of the proximal box. Research comparing different insertion and light-curing techniques is not conclusive and no single technique has been universally accepted. The number of increments will depend on the size of the proximal box. At the University of North Carolina, we recommend an oblique incremental technique: The first increment(s) should be placed along the gingival floor and should extend slightly up the facial (or lingual) wall (see Fig. 8.23). This increment (or increments for a large box) should be no more than 2 mm thick because it is the farthest increment from the curing light and the most critical in establishing a proper gingival seal. A second increment is then placed against the lingual (or facial) wall to restore about two thirds of the box. The final increment is then placed to complete the proximal box and develop the marginal ridge. Subsequent additions, if needed, are made and light cured (usually not exceeding 2 mm in thickness at a time) until the proximal box is fully restored (see Fig. 8.23). Increments should be light cured for as long as needed, depending on the shade and opacity of the composite used, the distance of the composite from the light tip, and the power of the light-curing unit. Regardless of the number of increments needed, when restoring the proximal box, an effort should be made to develop the anatomy of marginal ridge without excessive composite, to reduce the amount of rotary instrumentation required during finishing.



• **Fig. 8.23** Oblique incremental technique to restore proximal boxes in Class II direct composite restorations. The number of increments will depend on the size of the proximal box. The first increment(s) *(1)* should be placed along the gingival floor and should extend slightly up the facial wall. This increment should be only approximately 1 to 2 mm thick because it is the farthest increment from the curing light and the most critical in establishing a proper gingival seal. A second increment *(2)* is then placed against the lingual wall, to restore about two thirds of the box. The final increment *(3)* is then placed to complete the proximal box and develop the marginal ridge. Subsequent additions, if needed, are made and light cured (usually not exceeding 2 mm in thickness at a time) until the proximal box is filled.

When the proximal box is completed, the occlusal step of the preparation is restored exactly as it was described for the Class I direct composite restoration—that is, using an anatomic layering technique.

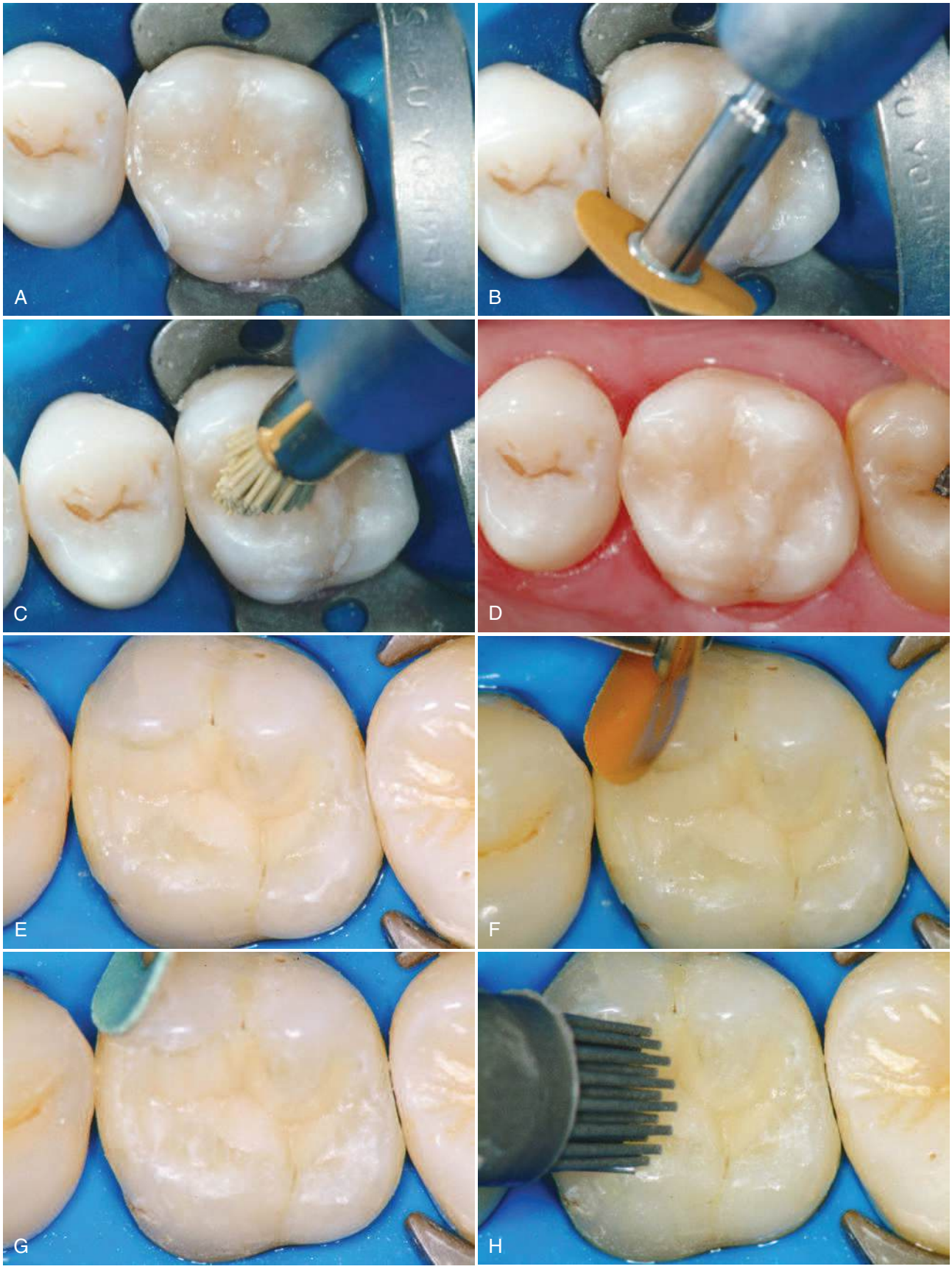
The incremental insertion and light-curing technique described provides enhanced control over the application and polymerization of individual increments of composite. The incremental technique also allows for (1) orientation of the polymerization light beam according to the position of each increment of composite, thus enhancing the polymerization potential; (2) intrinsic restoration characterization with darker or pigmented composites; and (3) sculpture of the restoration occlusal stratum with a more translucent material simulating the natural enamel. Appropriate proximal contact intensity can also be better achieved when composite is applied in increments. The matrix can be held in physical contact with the adjacent proximal surface while the contact-related increment of composite is light cured. A hand instrument with a large surface area (e.g., a small football-shaped or round burnisher) is well suited for that purpose. Once this increment is cured, the proximal contact is established, and remaining increments can be inserted and light cured. The matrix is removed and the restoration is light cured from the facial and lingual directions. The restoration is finished and polished immediately after the last increment is cured.

Composite resin placement may be made more difficult by the stiffness and stickiness of some composite materials. Heating the composite material prior to insertion in the preparation may help overcome these problems. Commercial "composite warmers" are available (e.g., Calset, AdDent Inc., Danbury, CT) to preheat the composite resin to preset temperatures up to 68°C (155°F). The increased temperature lowers the viscosity of the composite resin, potentially resulting in better marginal adaptation and reduced microleakage, although some of these results have been shown to be composite specific.[46-49] The elevated composite temperatures have been shown to be safe for clinical use.[50,51]

When a stiffer or "packable" high-viscosity composite is used for the restoration of the proximal box, a very small increment of a flowable composite may be first placed in close proximity to the external margins of the proximal box so as to improve marginal adaptation of the restoration.[52,53]

Finishing and Polishing of the Composite

Finishing can be initiated immediately after the composite material has been fully light cured (Fig. 8.24). If the occlusal anatomy



• **Fig. 8.24** Contouring and finishing of posterior composite restorations. A, After sectional matrix removal, excess composite is noted on facial and lingual proximal embrasures. B, Contouring lingual embrasure with finishing disk. C, Polishing the occlusal surface with finishing brush. D, Completed restoration. E, Note minimal excess composite after removal of sectional matrix because of good matrix adaptation to facial and lingual embrasures and careful incremental insertion of the composite. F–H, Contouring and polishing with disks and brush.

was developed as described in the previous sections, the need for additional contouring is greatly minimized. If finishing is needed, the occlusal surface is shaped with a round or oval, 12-bladed carbide finishing bur or finishing diamond. Excess composite is removed at the proximal margins and embrasures with a flame-shaped, 12-bladed carbide finishing bur or finishing diamond and abrasive discs (see Fig. 8.24B). Any overhangs at the gingival area are removed with a No. 12 surgical blade mounted in a Bard-Parker handle with light shaving strokes to remove the excess. Narrow finishing strips may be used to smooth the gingival proximal surface. Care must be exercised in maintaining the position of the finishing strips gingival to the proximal contact area to avoid inadvertent removal of the composite contact with the adjacent proximal surface. The rubber dam (or other means of isolation) is removed, and the occlusion is evaluated for proper contact. Further adjustments are made, if needed, and the restorations are polished with appropriate polishing points, cups, brushes, or discs (see Fig. 8.24C, F–H).

Extensive Class II Direct Composite Restorations and Foundations

Direct composite is not usually indicated for extensive posterior restorations but may be used as such when economic factors prevent the patient from selecting a more ideal (and more costly) indirect restoration, or as a semipermanent restoration in teeth with questionable prognosis. The 12-year survival rate of large posterior composite restorations has been shown to be similar to that of large amalgam restorations, although amalgam performed better in patients with high caries risk.[54] In addition to being used in selected cases for extensive restorations, composites also may be considered for use as a foundation for indirect restorations (crowns and onlays) when the operator determines that insufficient natural tooth structure remains to provide adequate retention and resistance form for the crown. The tooth first is restored with a large restoration and is then prepared for the indirect restoration.

In addition to the tooth preparation form, the primary retention form for a very large Class II composite restoration is the micromechanical bonding of the composite to enamel and dentin. When a full-coverage preparation is anticipated, secondary retention features must be incorporated, however, because of (1) the decreased amount of tooth structure available for bonding and (2) the increased concern for retaining the composite in the tooth. These features may include grooves, coves, and slots.

The primary differences for these very large preparations include the following: (1) Some or all of the cusps may be reduced and covered with composite ("capped"), (2) extensions in most directions are greater than in more conservative preparations, (3) secondary retention features are used, and (4) additional resistance form features are used. A cusp must be reduced and covered with composite if loss of supporting dentin is severe enough to leave the cusp in a weakened, fracture prone state. Reducing and covering a cusp with composite usually is indicated when the occlusal outline form extends more than two thirds the distance from a primary groove to a cusp tip. An operator sometimes may choose to ignore this general rule when using a bonded restoration if that weakened cusp will be covered and reinforced as part of the preparation design for the subsequent indirect restoration.

If the tooth has had endodontic treatment, the pulp chamber can be opened and extensions can be made several millimeters into each treated canal. Because of the increased surface area for bonding and the mechanical retention from extensions into the canals, usually fewer secondary retention features are incorporated into the remaining tooth preparation.

The elongated pear-shaped diamond or bur is used to prepare the occlusal step. As already noted, the occlusal outline form is usually extensive. When moving the instrument from the central groove area toward a cuspal prominence, the pulpal depth that is approximately 0.2 mm inside the DEJ should be maintained, if possible. This creates a pulpal floor, which rises occlusally as it is extended either facially or lingually (see Fig. 8.8). If a cusp must be reduced and covered, the side of the rotary instrument can be used first to make several depth cuts in the remaining cuspal form to serve as a guide for cusp reduction. Cusps should be reduced as early in the tooth preparation procedure as possible, providing more access and visibility for the preparation. The depth cut is made with the instrument held parallel to the cuspal incline (from cusp tip to central groove) and approximately 1.5 to 2 mm deep. For a large cusp, multiple depth cuts can be made. Then, the instrument is used to join the depth cuts and extend to the remainder of the cuspal form (Fig. 8.25). The reduced cusp has a relatively flat surface that may rise and fall with the normal mesial and distal inclines of the cusp. It also should provide enough clearance with the opposing tooth to result in approximately 1.5 to 2 mm of composite material to restore form and function. The cusp reduction should be blended in with the rest of the occlusal step portion of the preparation.

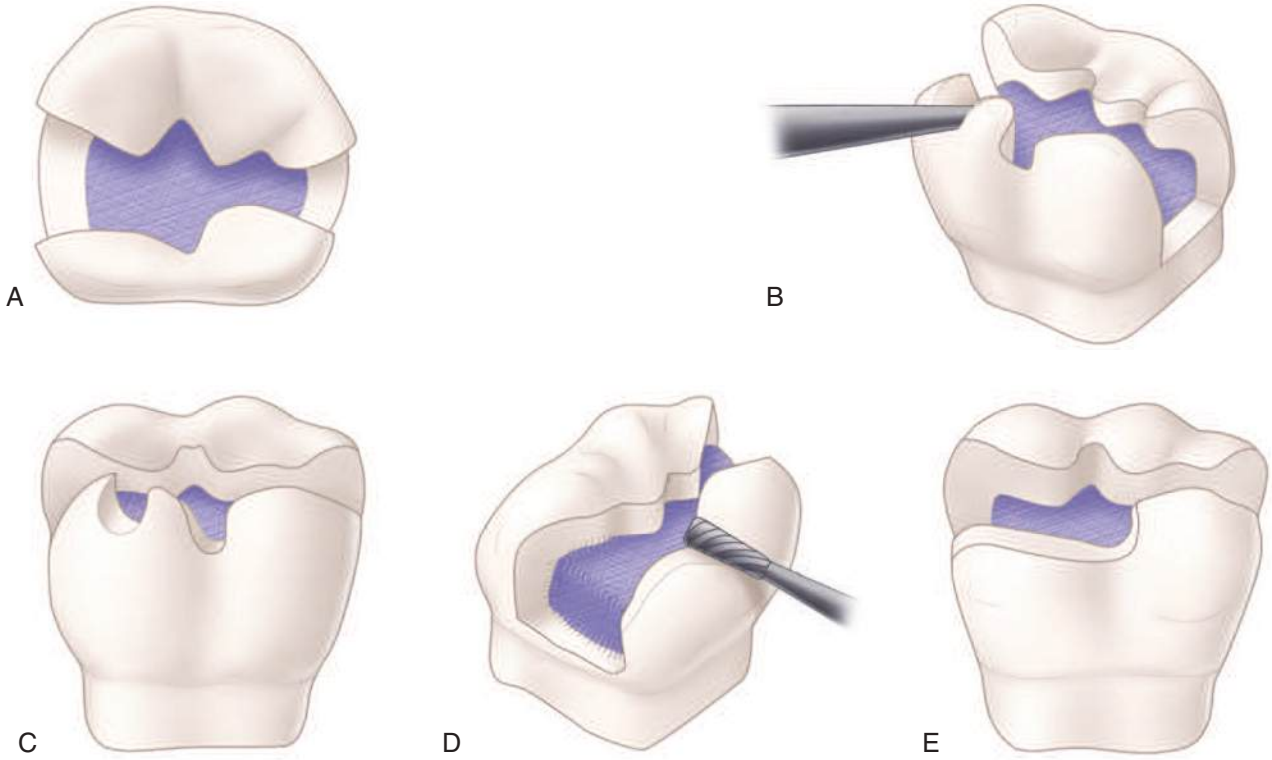
The proximal boxes are prepared as described previously. The primary difference is that they may be much larger, that is, extend farther in every direction. The extent of the caries lesion may dictate that a proximal box extend around the line angle of the tooth to include the caries lesion or faulty facial or lingual tooth structure. When the outline form has been established (the margins extended to sound tooth structure), carious tissue at the pulpal and axial walls is selectively removed and the preparation is assessed carefully for additional retention form needs.

Retention form can be enhanced by the placement of grooves, coves, or slots. All such retention form features must be strategically placed in dentin so as to maintain dentinal support of remaining enamel while avoiding areas that increase the risk of pulpal involvement. At times, bevels may be placed on available enamel margins to enhance retention form, even on occlusal areas. Retention form for foundations must be placed far enough inside the DEJ (at least 1 mm) to remain after the crown preparation is subsequently accomplished. Otherwise, essential secondary features that are retaining the foundation may potentially be lost (Fig. 8.26).

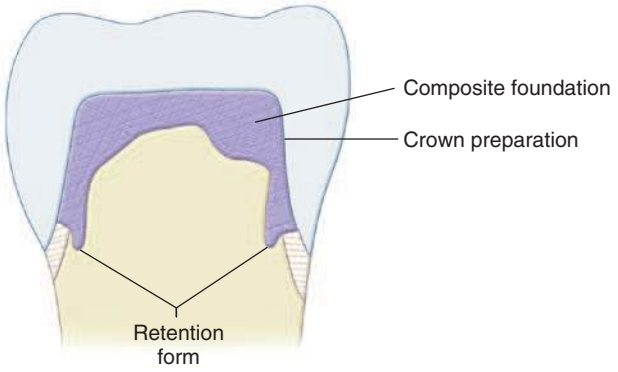
Matrix placement is more demanding for these large restorations because more tooth structure is missing, and more margins may be subgingival. Proper burnishing of the matrix band to achieve appropriate axial contours is important, unless immediate full coverage of the tooth is planned. It also may be necessary to modify the matrix band to provide more subgingival extension in some areas and prevent extrusion of the composite at the matrix band–retainer tooth junction.

Typical adhesive placement techniques are followed. Because much of the composite bond is to dentin, proper technique is crucial. Placement of the adhesive is accomplished as previously described.

When a light-cured composite is used, first it is placed in 1- to 2-mm increments into the most gingival areas of the proximal boxes. Each increment is cured, as directed. It may be helpful to use a hand instrument to hold the matrix against the adjacent tooth while light curing the composite. This may assist in restoring the proximal contact.



• **Fig. 8.25** Cusp reduction. A, The initial outline form weakens the mesiolingual cusp enough to necessitate capping. B, Depth cuts made. C, Depth cuts. D, Cusp reduction prepared. E, Vertical wall maintained between reduced and unreduced cusps.



• **Fig. 8.26** The retention form for foundations must be internal to eventual crown preparation.

Self-cured and dual-cured composite resin materials are frequently used for large *composite foundations* because these can be injected in the preparation in a single increment. However, it is recommended that even when dual-cured composites are used, they be carefully light cured during and after the final placement. When this technique is used, the operator should carefully select the adhesive system, as some simplified adhesives have been shown to be incompatible with some self-cured composite foundation materials. Acidic monomers in these adhesives scavenge the activators (tertiary amines) in the self-cure composite.[55,56] If the activator does not function properly, then the composite at the adhesive interface does not polymerize thoroughly and does not polymerize with the previously placed adhesive. Some manufacturers have introduced optional chemical catalysts that can be mixed with the light-cured adhesive to reduce or prevent this problem.

Contouring the large composite also is more difficult because of the number of surfaces that may be involved and the amount of composite present. Anatomically correct contours and contacts are necessary for restorations, but they are less important for foundations that will be present only for a short time before the crown preparation is accomplished and the tooth is stabilized. Because of the extensiveness of these restorations, a careful assessment of the contours should be made from all angles. When finishing has been completed, the occlusion is adjusted as necessary, and the restoration is polished. Because these very large restorations may stretch the limits of composite restorations, the patient should be on a frequent recall regimen.

Clinical Technique for Class III Direct Composite Restorations

Initial Clinical Procedures

These are similar to those already described: (1) Anesthesia is usually necessary for patient comfort and may help decrease salivary flow during the procedure; (2) occlusal assessments should be made to determine the tooth preparation design and to properly adjust the restoration's function; (3) the composite shade must be selected before the tooth dehydrates and concomitantly lightens; (4) the area must be isolated to facilitate access and permit effective bonding;

and (5) if the restoration involves the proximal contact, inserting a wedge in the area beforehand may assist in the reestablishment of the proximal contact with composite.

Tooth Preparation

In general, the tooth preparation for a Class III direct composite restoration involves (1) obtaining access to the defect (caries lesion, fracture, noncarious defect), (2) removing faulty structures (carious tissue, defective dentin and enamel, defective restoration, base material), and (3) creating the convenience form for the restoration (Fig. 8.27). In most cases, an enamel bevel is used on the facial cavosurface margin to provide a gradual color transition from the restoration to the surrounding tooth structure for esthetics. If unsupported enamel is intentionally left on the facial aspect of the preparation, a facial cavosurface bevel may not be indicated as it can unnecessarily extend the facial display of the restoration. Obtaining access to the defect may include removal of sound enamel to access the caries lesion. The extension of the preparation is therefore ultimately dictated by the extension of the fault or defect. It is usually not necessary to reduce sound tooth structure to provide either "bulk for strength" or conventional retention and resistance forms.

Because of the adequate bond of composite to enamel and dentin, most Class III composite restorations are retained almost exclusively by bonding, and no additional preparation retention form is necessary. In the rare cases when additional retention form is needed, it can be achieved either by increasing the surface area with a wider enamel bevel or by adding retentive features in the preparation internal dentin walls.

When a proximal surface of an anterior tooth is to be restored, and a choice between facial or lingual entry into the tooth is available, the lingual approach is always preferable unless such an approach would necessitate excessive removal of the tooth structure, such as in instances of irregular alignment of teeth or facial positioning of the lesion. The advantages of restoring the proximal lesion using a *lingual approach* are:

1. The facial enamel is conserved for enhanced esthetics. (Some unsupported, but nonfriable, enamel may be left on the facial wall of the preparation.)
2. Shade matching of the composite is less critical.
3. Discoloration or deterioration of the restoration is less visible.

Indications for a *facial approach* include:

1. The caries lesion is positioned facially, and facial access would significantly conserve the tooth structure.
2. Teeth are irregularly aligned, and facial access would significantly conserve the tooth structure.
3. An extensive caries lesion extends onto the facial surface.
4. A faulty restoration that originally was placed from the facial approach needs to be replaced.

When the facial and the lingual surfaces are involved, the approach that provides the best access for instrumentation should be used.

The preparation is initiated from a lingual approach (if possible) by using a round carbide bur or diamond instrument of a size compatible with the extent of the lesion. The point of entry is located within the incisogingival dimension of the lesion or defect and as close to the adjacent tooth as possible without contacting it (Fig. 8.28A). The initial opening is made by holding the cutting instrument perpendicular to the enamel surface but at an entry angle that places the neck portion of the bur or diamond instrument as far into the embrasure (next to the adjacent tooth) as possible; light pressure and intermittent cutting (brush stroke) are used to gain access into the preparation. Incorrect entry overextends the lingual outline and unnecessarily weakens the tooth (see Fig. 8.28B and C).

The same instrument may be used to enlarge the initial access opening sufficiently to permit, in subsequent steps, carious tissue removal, completion of the preparation, and insertion of the restorative material (see Fig. 8.28D). No effort is made to prepare the walls that are perpendicular to the enamel surface; for small preparations, the walls may diverge externally from the axial depth resulting in a beveled marginal design and conservation of internal tooth structure (Fig. 8.29). For larger preparations, the initial tooth preparation still is as conservative as possible, but the preparation walls may not be as divergent from the axial wall. Subsequent beveling or flaring of accessible enamel areas may be required. Despite the size of the lesion, the objective of the initial tooth preparation is the same: to prepare the tooth as conservatively as possible by extending the outline form just enough to include the peripheral extent of the lesion. Sometimes, the incorporation of an enamel bevel also may be used to extend the final outline form to include the caries lesion. If possible, the outline form should not (1) include the entire proximal contact area, (2) extend onto the facial surface, or (3) extend subgingivally. Extensions should be minimal, including only the tooth structure that is compromised by the extent of the caries lesion or defect. Some undermined enamel can be left in nonocclusal stress areas, but very friable enamel at the margins should be removed.

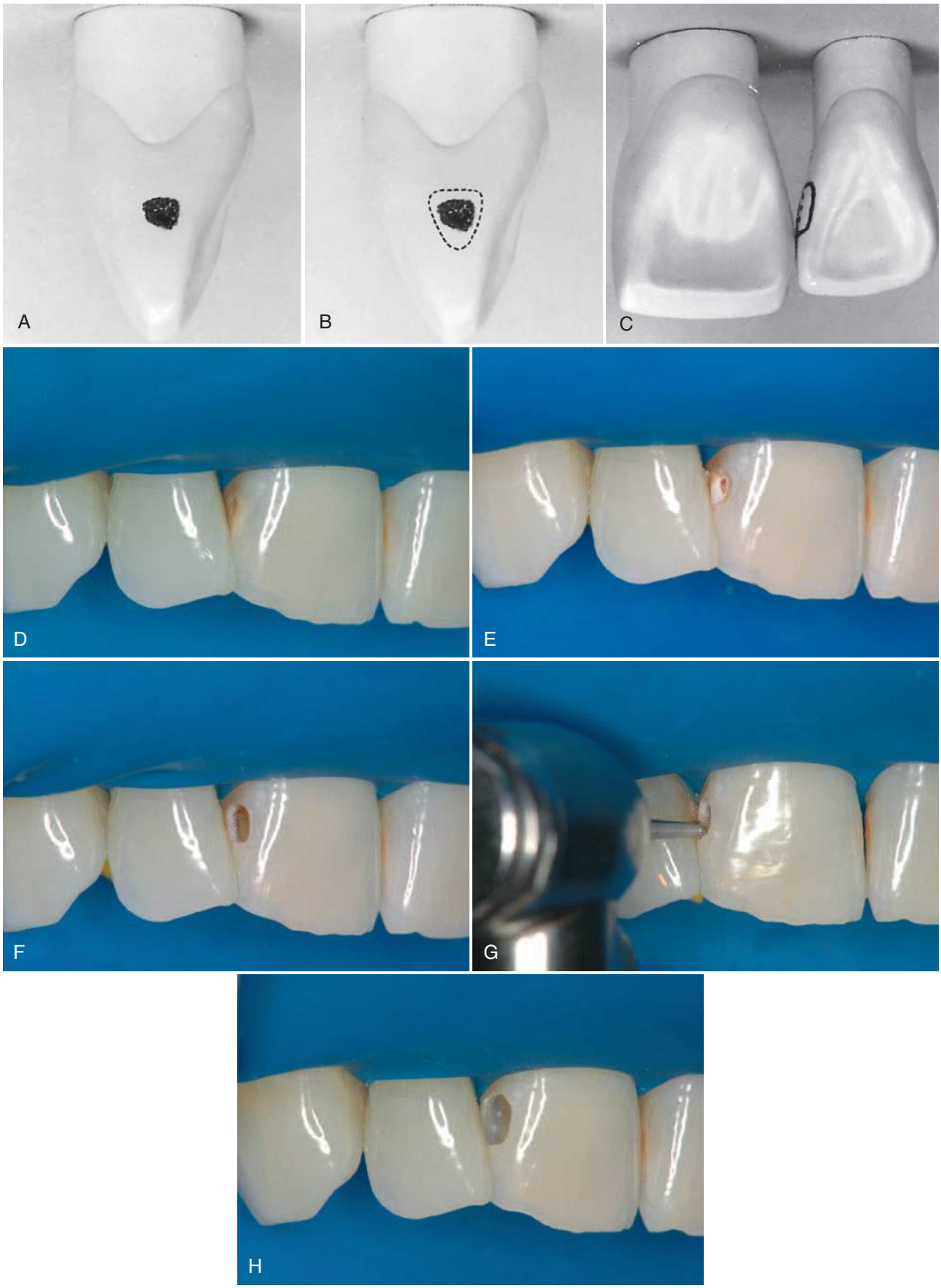
The extension axially also is dictated by the extent of the fault or caries lesion and usually is not uniform in depth. As noted earlier, most initial composite restorations (primary caries) use an ultraconservative preparation design (Fig. 8.30A and B). Because a caries lesion that requires a restoration usually extends into dentin, many Class III preparations extend to an initial axial wall depth of 0.2 mm into dentin and are widened such that peripheral enamel has dentinal support (Fig. 8.31). No attempt is made however to prepare distinct or uniform axial preparation walls; rather, the objective is to selectively remove carious tissue as conservatively as possible. Additional marginal refinement may be necessary later.

If the preparation outline extends gingivally onto the root surface, the gingival floor should form a cavosurface margin of 90 degrees, and the depth of the gingivoaxial line angle should be not more than 0.75 mm at this initial stage of tooth preparation. The external walls are prepared perpendicular to the root surface. In this area of the tooth, apical of the cementoenamel junction (CEJ), the external walls are composed entirely of dentin and cementum.

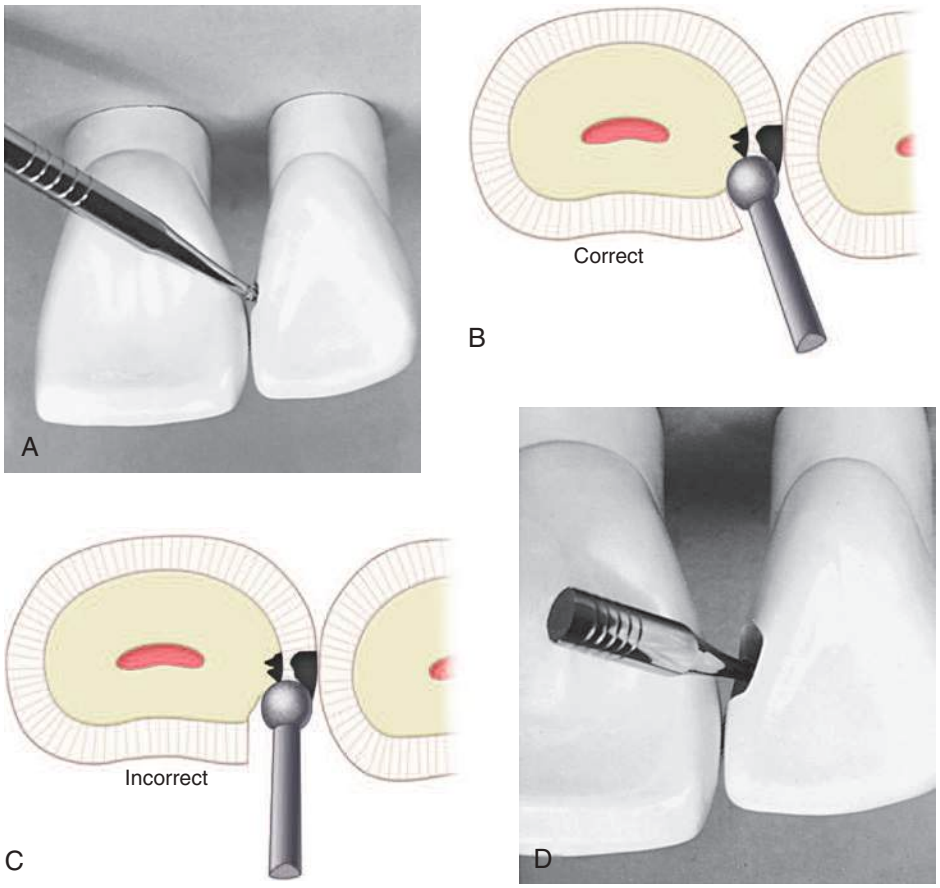
When completed, the initial tooth preparation extends the outline form to include the entire fault unless it is anticipated that the incorporation of an additional enamel bevel would enhance retention form or improve the potential for an optimal esthetic result. Small preparations typically have a beveled marginal configuration from the initial tooth preparation.

Little may need to be done in the final tooth preparation stage for these preparations. Final tooth preparation steps for a Class III direct composite restoration are, when indicated, (1) selective removal of carious dentin; (2) pulp protection; (3) bevel placement on accessible enamel margins; and (4) final procedures of cleaning and inspecting. The selective removal of carious dentin is accomplished using round burs, small spoon excavators, or both. Particular care must be exercised not to weaken the walls or incisal angles that are subject to masticatory forces.

Larger preparations may require additional beveling of the accessible enamel walls to enhance retention by bonding (Fig.



• **Fig. 8.27** A, Small proximal caries lesion on the mesial surface of a maxillary lateral incisor. B, Dotted line indicates normal outline form dictated by shape of the caries lesion. C, Extension (convenience form) required for preparing and restoring preparation from lingual approach when teeth are in normal alignment. D–H, Clinical case showing conservative Class III preparation, facial approach. D, Facial view of a caries lesion on the distal surface of the maxillary central incisor. E and F, Obtaining access to carious dentin. G, Infected dentin is removed with round bur. H, Completed caries excavation.



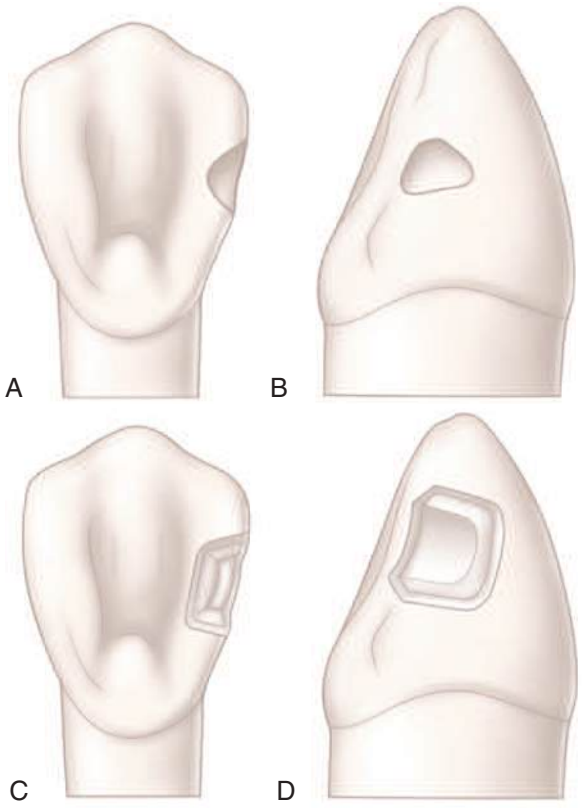
• **Fig. 8.28** Beginning Class III tooth preparation (lingual approach). A, The bur or diamond is held perpendicular to the enamel surface, and an initial opening is made close to the adjacent tooth at the incisogingival level of the caries. B, The correct angle of entry is parallel to the enamel rods on the mesiolingual angle of the tooth. C, Incorrect entry overextends the lingual outline. D, The same bur or diamond is used to enlarge opening for caries removal and convenience form while establishing the initial axial wall depth.



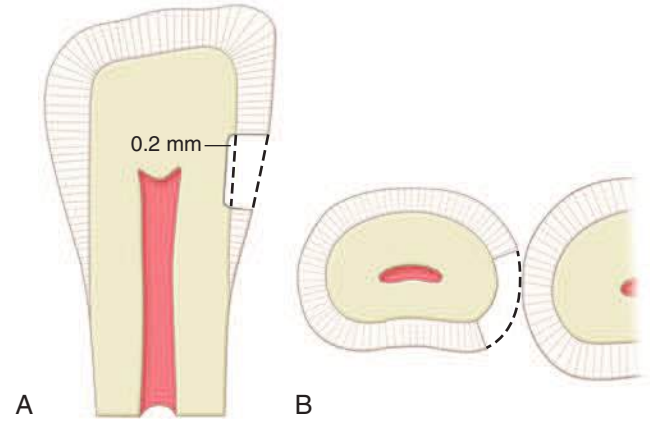
• **Fig. 8.29** A small, scoop-shaped Class III tooth preparation.

8.32; see Fig. 8.30C and D). These enamel margins are beveled with a flame-shaped or round diamond instrument. The bevel is prepared by creating a 45-degree angle to the external surface and to a width of 0.5 to 2 mm, depending on the size of the preparation, location of the margin, and esthetic requirements of the restoration (Fig. 8.33; see Fig. 8.32). If the gingival floor has been extended gingivally to a position where the remaining enamel thickness is minimal or nonexistent, the bevel is omitted from this area to preserve the remaining enamel margin or maintain a 90-degree cavosurface margin in dentin. Likewise, a bevel on the lingual enamel margin of a maxillary incisor may be contraindicated if it would result in the creation of a restoration marginal interface at the area of occlusal contact.

Remaining old restorative material on the axial wall should be removed if any of the following conditions are present: (1) The old material is amalgam, and its color would negatively affect the color of the new restoration; (2) clinical or radiographic evidence of a caries lesion under the old material is present; (3) the tooth pulp was symptomatic preoperatively; (4) the periphery of the remaining restorative material is not intact; (5) there is noticeable deterioration of the bonding of the remaining restorative material; or (6) the use of the underlying dentin is necessary to effect a stronger bond for retention purposes. If none of these conditions is present, the operator may elect to leave the remaining restorative material rather than risk unnecessary excavation nearer to the pulp and subsequent irritation or exposure of the pulp. A RMGI base is applied only if the remaining dentin thickness is judged to be less than 1.5 mm and in the deepest portions of the preparation.[26] Calcium hydroxide liners are used only in cases when the remaining dentin thickness is 0.5 mm or less as an indirect pulp-capping



• **Fig. 8.30** Preparation designs for Class III (A and B); larger preparation designs for Class III (C and D).

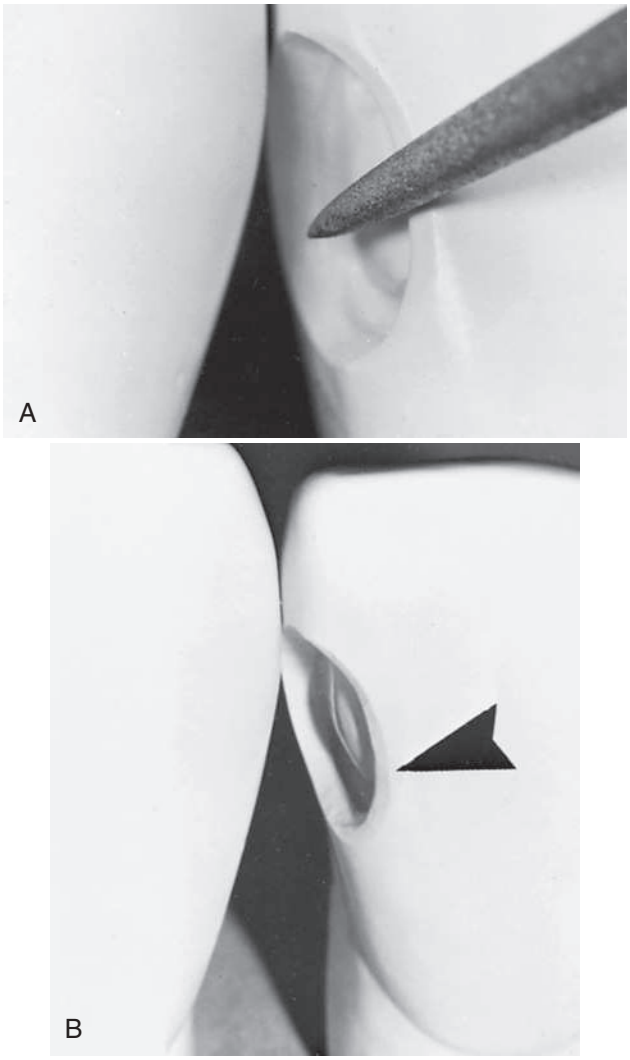


• **Fig. 8.31** Ideal initial axial wall preparation depth. A, Incisogingival section showing axial wall 0.2 mm into dentin. B, Faciolingual section showing facial extension and axial wall following the contour of the tooth.

material; in cases of pulp exposures calcium hydroxide or, increasingly, MTA can be used as a direct pulp-capping material.[26-28] If used, the calcium hydroxide or MTA liners should always be covered with a RMGI base, sealing the area and preventing the etchant (applied later) from dissolving the liner.[26,29]

When replacing a large restoration or restoring a large Class III lesion, the operator may decide that retention form should be enhanced by placing groove (at gingival) or cove (at incisal) retention features in addition to bonding.

For Class III direct composite preparations with *facial access*, with a few exceptions, the same stages and steps of tooth preparation

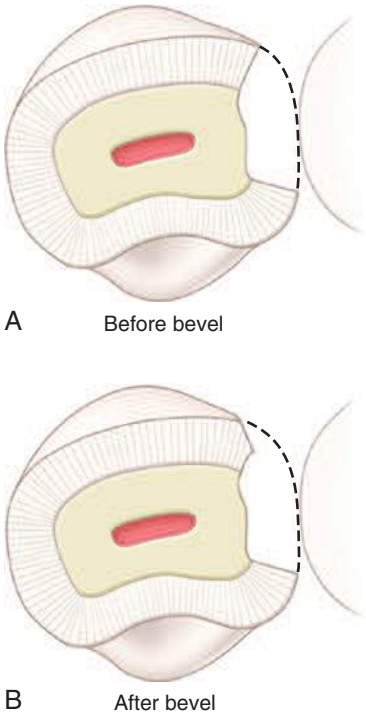


• **Fig. 8.32** Large Class III tooth preparation. A, Beveling. The cavosurface bevel is prepared with a flame-shaped or round diamond, resulting in an angle approximately 45 degrees to the external tooth surface. B, Completed cavosurface bevel *(arrowhead)*.

are followed as for lingual access. The ability to use direct vision simplifies the procedure (Fig. 8.34).

It is expeditious to prepare and restore approximating caries lesions or faulty restorations on adjacent teeth at the same appointment. If one of the lesions is larger (more extended outline form) than the other, then the larger outline form is developed first. The second preparation usually can be more conservative because of the improved access provided by the larger preparation. The reverse order would be followed when the restorative material is inserted.

A Class III lesion on the distal surface of a maxillary right central incisor is shown in Fig. 8.35A. The rubber dam is placed after the anesthetic has been administered and the shade has been selected. A wedge is inserted in the gingival embrasure to depress the rubber dam and underlying soft tissue, improving gingival access (see Fig. 8.35B). Using a carbide bur or diamond instrument rotating at high speed and with air-water spray, the outline form is prepared with appropriate extension and the initial, limited pulpal depth as previously described in the lingual approach preparation (see Fig. 8.35C). Carious tissue is removed with a



• **Fig. 8.33** Cross section of facial approach Class III before (A) and after (B) 45-degree cavosurface bevel on the facial margin.

spoon excavator (see Fig. 8.35D). Some undermined enamel may be left if it is not in a high-stress area.

When a proximal caries lesion or defective restoration extends onto the facial *and* the lingual surfaces, access may be accomplished from either a facial approach or a lingual approach. An example of an extensive Class III initial tooth preparation that allows such choice is illustrated in Fig. 8.36.

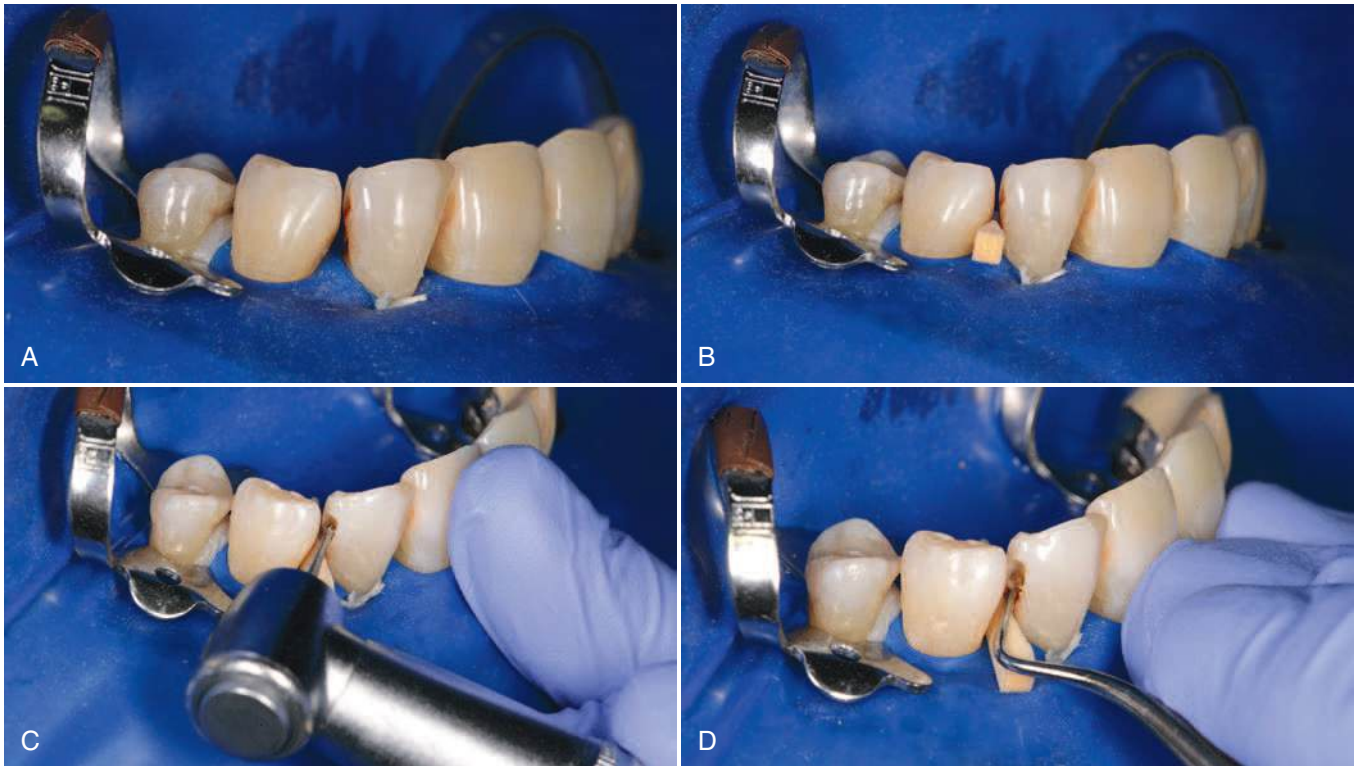
Restorative Technique

Matrix Application

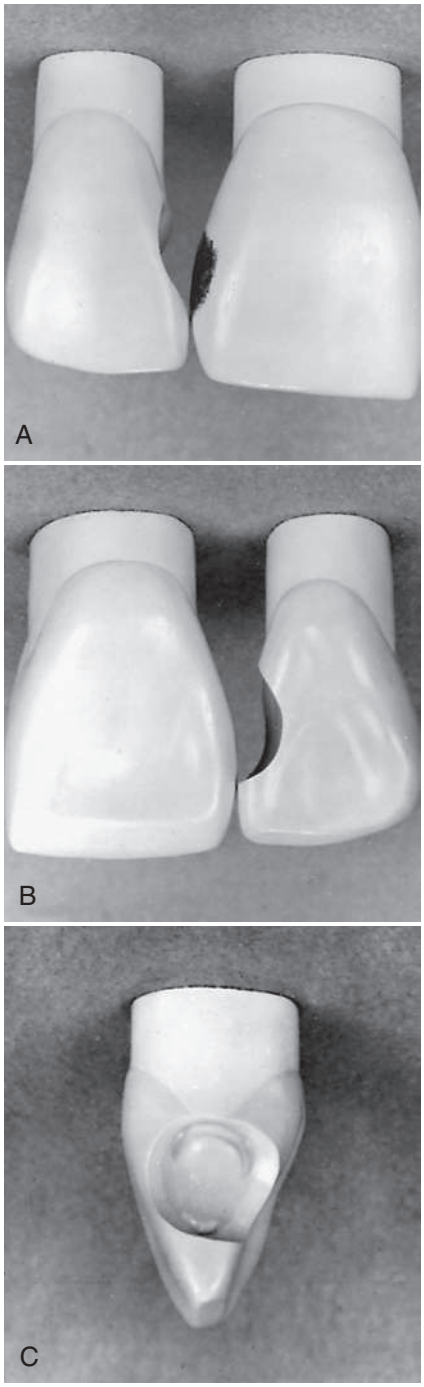
A matrix is a device that is applied to a prepared tooth before the insertion of the restorative material to (1) confine the restorative



• **Fig. 8.34** Completed Class III tooth preparation (facial approach), with the bevel marked.



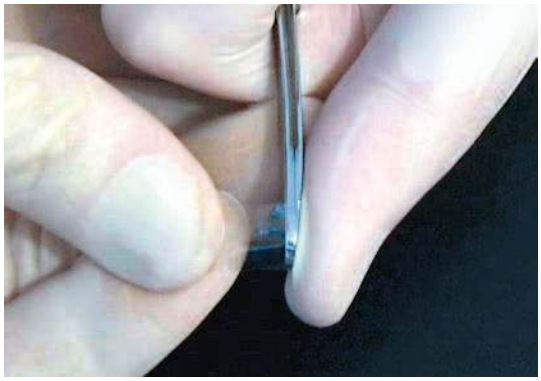
• **Fig. 8.35** Class III initial preparation (facial approach). A, Large proximal caries with facial involvement. B, Isolated area of operation. C, Entry and extension with No. 2 bur or diamond. D, Caries removal with spoon excavator. (Courtesy Vilhelm G. Ólafsson.)



• **Fig. 8.36** Large Class III tooth preparation extending onto root surface. A, Facial view. B, Lingual view. C, Mesial view showing gingival and incisal retention, which is only used when deemed necessary to increase retention. The tooth preparation is now ready for beveling of the enamel walls.

material excess and (2) assist in the development of the appropriate axial tooth contours. The matrix usually is applied and stabilized with a wedge before application of the adhesive because it helps contain the adhesive components to the prepared tooth. Care must be taken, however, to avoid pooling of adhesive adjacent to the matrix.

A properly contoured and wedged matrix is a prerequisite for a restoration involving the entire proximal contact area, unless the adjacent tooth is missing in which case the restoration may be completed by direct placement of the composite. When correctly used, not only would a matrix aid in placing and contouring the composite restorative material, but it may also reduce the amount of excess material, thus minimizing the finishing time.



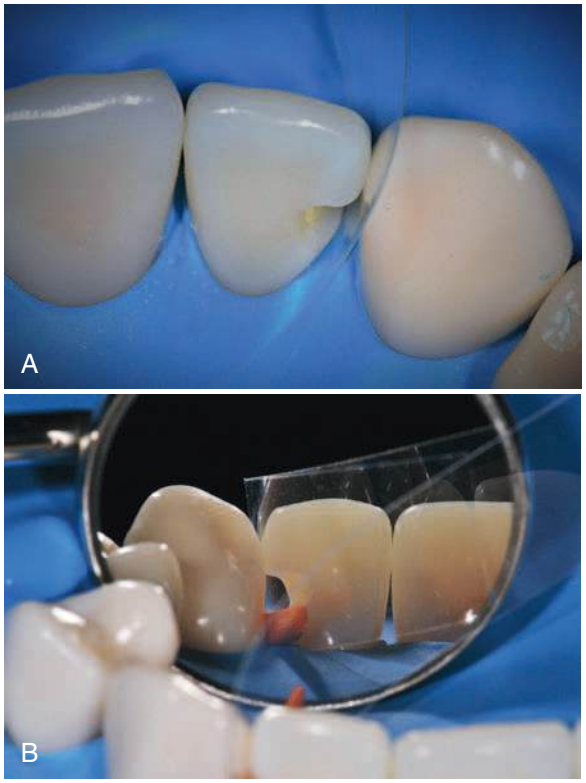
• **Fig. 8.37** Contouring Mylar strip matrix. (Courtesy Aldridge D. Wilder, DDS.)

A properly contoured section of Mylar matrix material is used for most Class III and IV preparations. Because the proximal surface of a tooth is usually convex incisogingivally and the matrix material may be flat, it is necessary to shape the matrix to conform to the desired tooth contour. One way to contour a Mylar matrix is by drawing it along a hard, rounded object (Fig. 8.37). The amount of convexity placed in the matrix depends on the size and contour of the anticipated restoration. Several pulls of the matrix, with heavy pressure, across the rounded end of the operating pliers may be required to obtain enough convexity. The contoured matrix is positioned between teeth so that the convex area conforms to the desired tooth contour (Fig. 8.38A). The matrix is extended at least 1 mm beyond the prepared gingival and incisal margins. Sometimes, the matrix does not slide through or is distorted by a tight contact or preparation margin. In such instances, a wedge is lightly positioned in the gingival embrasure before the matrix is inserted. Care must be taken not to injure the interproximal tissues and induce bleeding. When the matrix is past the binding area, it may be necessary to loosen the wedge to place the strip past the gingival margin (between the wedge and margin). Then the wedge is reinserted tightly (see Fig. 8.38B).

A wedge is needed at the gingival margin to (1) hold the Mylar matrix in position, (2) provide slight separation of the teeth, and (3) prevent a gingival overhang of the composite material. A wedge must be used to separate teeth sufficiently to compensate for the thickness of the matrix if the completed restoration is to contact the adjacent tooth properly.

Several types of commercial wedges are available in assorted sizes. A triangular-shaped wedge (in cross section) is indicated for preparations with margins that are deep in the gingival sulcus. The end of a round wooden toothpick usually is an excellent wedge for preparations with margins coronal to the gingival sulcus. The wedge is kept as short as possible so as not to limit operator access to the preparation during insertion of the restorative material.

The wedge is placed, using No. 110 pliers, from the facial approach for lingual access preparations, and vice versa for facial access, just apical to the gingival margin. When isolation is accomplished with the rubber dam, wedge placement may be aided by a small amount of water-soluble lubricant on the tip of the wedge. The rubber dam is first stretched gingivally (on the side from which the wedge is inserted), then released gradually



• **Fig. 8.38** Inserting and wedging Mylar strip matrix. A, Strip with concave area next to the preparation is positioned between teeth. B, Strip in position and wedge inserted. The length of the Mylar strip can be reduced as needed.

during wedge insertion (Fig. 8.39). Subsequently, a trial opening and closing of the Mylar matrix is helpful. It must open enough for access to insert the adhesive and composite and close sufficiently to ensure a proper contour. It may be necessary to shorten the wedge or insert it from the opposite embrasure to optimize access.

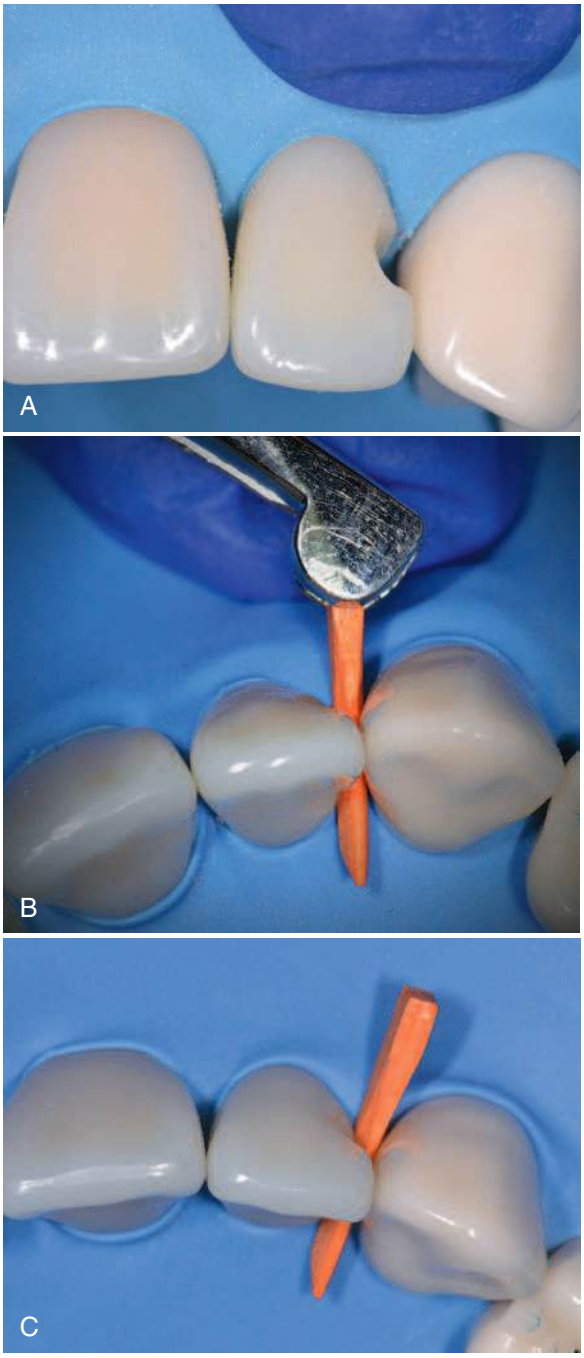
Placement of the Adhesive

Adhesive placement steps are accomplished with strict adherence to the manufacturer's directions for the particular adhesive system being used.

Insertion and Light Curing of the Composite

The mesial surface of a maxillary left lateral incisor is used to illustrate facial insertion of a light-cured composite (Fig. 8.40). The matrix is contoured, placed interproximally, and wedged at the gingival margin. The lingual aspect of the matrix is secured with the index finger, while the thumb reflects the facial portion out of the way (see Fig. 8.40). Light-cured materials are not dispensed until ready for use. The composite should be protected from ambient light to prevent premature polymerization. Likewise, the compule tip (when used) should be recapped immediately to prevent setting of the composite at the end of the syringe.

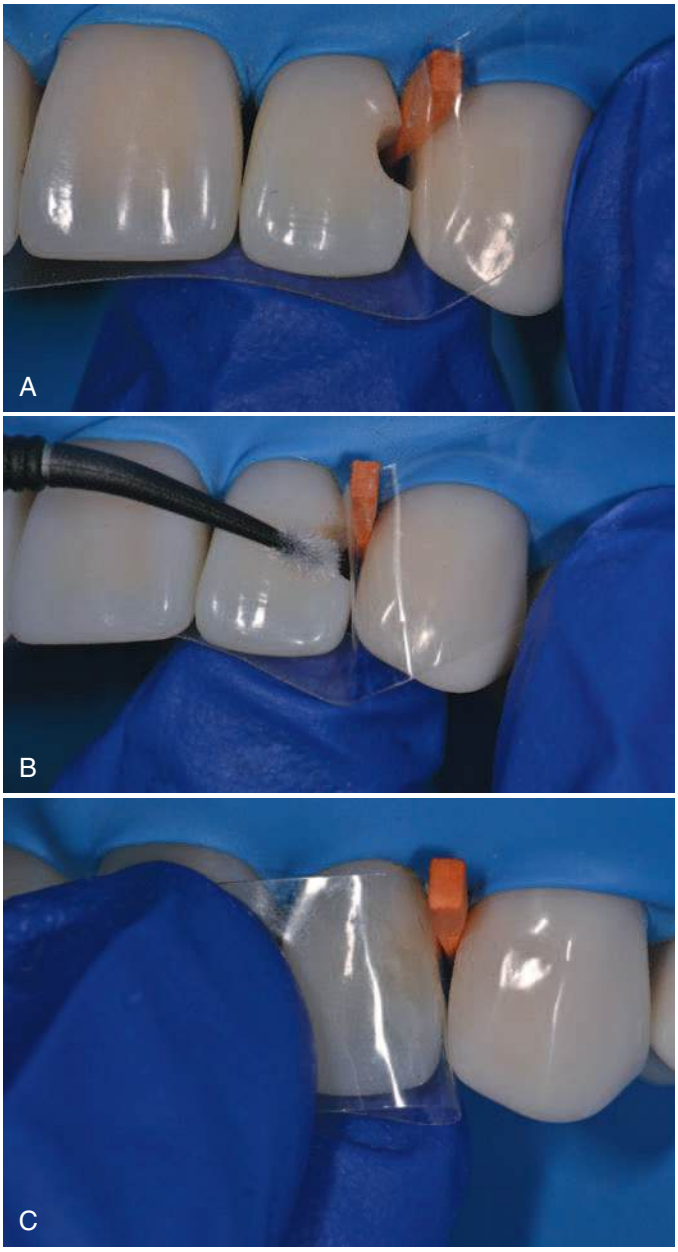
The composite is inserted by a hand instrument or syringe and pressed into the preparation. Most modern composites will not stick to a clean instrument, but if stickiness is perceived, the hand instrument should be thoroughly cleaned and dried. Once the desired aspect of the preparation is adequately restored, the material is polymerized. A second increment of composite is applied, if needed, to fill the preparation completely and provide a slight



• **Fig. 8.39** Using a triangular wood wedge to expose gingival margin of large proximal preparation. A, The dam is stretched facially and gingivally with the fingertip. B, Insertion of wedge (the dam is released during wedge insertion). C, Wedge in place.

excess so that positive pressure can be applied with the matrix strip when closed. Any gross excess is removed quickly with the blade of the insertion instrument or an explorer tine before closing the matrix.

The operator closes the lingual end of the matrix over the composite and holds it with the index finger. Next, the operator pulls the matrix toward the facial direction to ensure excellent adaptation of the composite with the facial margin. Before light curing the composite, the operator closes the facial end of the matrix over the tooth with the thumb and index finger of the



• **Fig. 8.40** Insertion of light-cured composite. A, Bonding adhesive is applied and light cured. B, The lingual aspect of the strip is secured with the index finger, while the facial portion is reflected away for access. C, After insertion of the composite, the matrix strip is closed and the material is cured through the strip.

other hand, tightening the gingival aspect of the matrix ahead of the incisal portion. The closed matrix is manually shifted so as to provide proximal contours that allow normal embrasure form and contact with the adjacent tooth, and the composite is light cured.

The composite is light cured through the matrix as directed (see Fig. 8.40C). The matrix should not be touched with the tip of the light initially because it could distort the contour of the restoration. The operator removes the index finger and light cures the lingual surface. Longer light exposures usually are required for the polymerization of dark and opaque shades. If the restoration is undercontoured, more composite can be added to the previously placed composite and light cured. No etching or adhesive is required between layers if the surface has not been contaminated, and the oxygen-inhibited layer remains.

With large restorations, it is better to add and light cure the composite in several increments to reduce the effects of polymerization shrinkage and to ensure complete light curing in remote regions. If two adjacent preparations are present, the preparation with the least access (usually the one prepared second) is restored first. Contouring of the proximal surface of the first restoration should be completed before the second restoration is started. Because the second tooth preparation has been contaminated, it must be cleaned (usually by use of the adhesive system etchant) before bonding materials and composite are applied. During these procedures, a Mylar matrix or piece of Teflon tape should be in place to protect adjacent surfaces.

Finishing and Polishing of the Composite

Good technique and experience in inserting composites significantly reduce the amount of finishing required. Usually a slight excess of material will need to be removed so as to provide correct contours. Similar to tooth preparation rotary instruments, finishing and polishing instruments should be used according to the specific surface being finished and polished. For example, flexible discs and finishing strips are suitable for convex and flat surfaces; oval-shaped finishing burs and polishing points are more suitable for concave surfaces; and polishing cups can be used in both convex and concave surfaces.

Coarse diamond instruments may be used to efficiently remove gross excess, but leave a rough surface on the restoration and the tooth. A flame-shaped carbide finishing bur or diamond is recommended for removing excess composite on facial surfaces. Medium speed with light intermittent brush strokes and air coolant are used to create correct contours. Special fine diamond finishing instruments, 12-bladed carbide finishing burs, and abrasive finishing disks may then be used to obtain excellent surface smoothness if the manufacturers' instructions are followed. Care must be exercised with all rotary instruments so as to prevent damage to the tooth structure, especially at the gingival marginal areas.

Abrasive discs (the degree of abrasiveness depends on the amount of excess to be removed) mounted on a mandrel specific to the disc type, in a contra-angle handpiece at low speed, can be used instead of or after the finishing bur or diamond in facial surfaces and some interproximal and incisal embrasures (Fig. 8.41A). Several brands of abrasive discs are available, and most are effective when used correctly. These discs are flexible and are produced in different diameters, thicknesses, and abrasive textures. Thin discs with small diameters fit into embrasure areas easily and are especially useful in finishing and polishing accessible embrasure areas. Regardless of the type of disc chosen, discs are used sequentially from coarse to very fine grit so as to generate a smooth, final surface. The external enamel surface should act as a guide for proper contour and surface texture. A constant shifting motion aids in contouring and preventing the development of a flat surface. Final polishing is initiated after correct contours and surface texture have been achieved and is accomplished with rubber or silicone polishing instruments, diamond-impregnated polishers, polishing discs, and polishing pastes.

Excess lingual composite is removed using a round or oval-shaped, 12-bladed carbide finishing bur or finishing diamond. A smoother surface is produced using a finer round or oval carbide finishing bur (with 18–24 or 30–40 blades) or fine diamond at medium speed with air coolant and light intermittent pressure



• **Fig. 8.41** Finishing composites. A, Abrasive disk mounted on mandrel can be used for finishing when access permits. B, The round carbide finishing bur is well suited for finishing lingual surfaces. C, The No. 12 surgical blade in Bard-Parker handle can be used for removing interproximal excess. D, The abrasive strip should be curved over the area to be finished. E and F, Replacement of old composite restoration, before (E) and after (F).

(see Fig. 8.41B). The appropriate size and shape depend on the amount of excess and shape of the lingual surface. Polishing is achieved with rubber polishing instruments and diamond-impregnated polishers.

Proximal surface contours and margins should be assessed visually and tactilely with an explorer and unwaxed, ultrathin dental floss. The floss is positioned apical to the gingival margin, adapted to the convex tooth surface, and slowly moved in an occlusal direction across the gingival tooth-restoration interface. The floss will begin to separate (catch, fray) as an indication of excess adhesive, composite, or other marginal defect. A No. 12 surgical blade mounted in a Bard-Parker handle (see Fig. 8.41C) is well suited for removing excess material from the gingival proximal area as it is thin and has a curved shape consistent with natural embrasure contours. The instrument should be moved from the tooth to the restoration or along the margins, using light shaving strokes, keeping a portion of the cutting edge on the external enamel surface as a guide to prevent overreduction. If a large amount of composite is removed with one stroke or in the wrong direction, it may compromise the restoration interface resulting in a marginal defect or void. Immediate correction of this area is indicated so as to limit the potential for plaque/debris accumulation resulting in additional caries and/or discoloration. Rotary instruments especially designed for this task are also available and can be used for removing excess and the creation of correct embrasure contours. Careless use of rotary instrumentation increases the risk of producing an undercontoured ("ledged") area just adjacent to the contact. All carbide instruments are made of carbon steel and may leave gray marks on the restoration. This discoloration is superficial and is removed easily during the final finishing by abrasive strips or discs (see Fig. 8.41D).

Further contouring of proximal surfaces may be completed with abrasive materials such as Sof-Lex Finishing Strips. These finishing strips have two different sizes of abrasive particles (e.g., medium and fine) impregnated on opposing ends of a polyester strip, with a small in-between area where no abrasive is present. The centrally located, nonabrasive area allows easy and safe passage of the strip through the contact area and apically to the gingival area in need of additional contouring. If the intensity of the contact is too great, then the finishing strip may need to be passed under the contact so as to position it for finishing the gingival/proximal contours. In addition, thin diamond-coated metal strips also are commercially available in various grits. Different widths of strips are available and the size of the strip must be selected based on the contouring needs of the particular situation. The strip should be curved over the restoration and tooth surface in a fashion similar to that used with a shoeshine cloth, concentrating on areas that need attention (see Fig. 8.41D). To open the lingual embrasure or round the marginal ridge, the lingual part of the strip is held against the composite with the index finger of one hand, while the other end of the strip is pulled facially with the other hand. Caution must be used so as to prevent gingival damage and/or excessive removal of the carefully created composite contours.

Finally, occlusion should be carefully checked after the rubber dam is removed, if one was used. The operator evaluates the occlusion in maximum intercuspation and eccentric movements by having the patient close on a piece of articulating paper and slide mandibular teeth over the restored area. If excess composite is present, the operator removes only a small amount at a time and rechecks with articulating paper. Usually, occlusion is adjusted until it does not differ from the original occlusion.

Instead of working sequentially by surface as previously described (facial, lingual, proximal, and occlusal), the operator may elect to work by instrument sequence—that is, contour all surfaces of the restoration by using finishing instruments first and then proceed to polish all surfaces by using the polishing instruments described. This approach improves efficiency of the overall procedure by limiting the number of times a finishing or polishing instrument is changed. Fig. 8.41E and F show before and after replacement of a Class III composite restoration.

Clinical Technique for Class IV Direct Composite Restorations

Initial Clinical Procedures

The same initial procedure considerations presented earlier are appropriate for Class IV direct composite restorations. The preoperative assessment of the occlusion is even more important for Class IV restorations because it might influence the tooth preparation extension (placing margins in noncontact areas) and retention and resistance form features (heavy occlusion requires increased retention and resistance form).

Also, proper shade selection can be more difficult for large Class IV restorations. Use of separate translucent and opaque shades of composite is often necessary. Specific information for esthetic considerations is presented in Chapters 7 and 9.

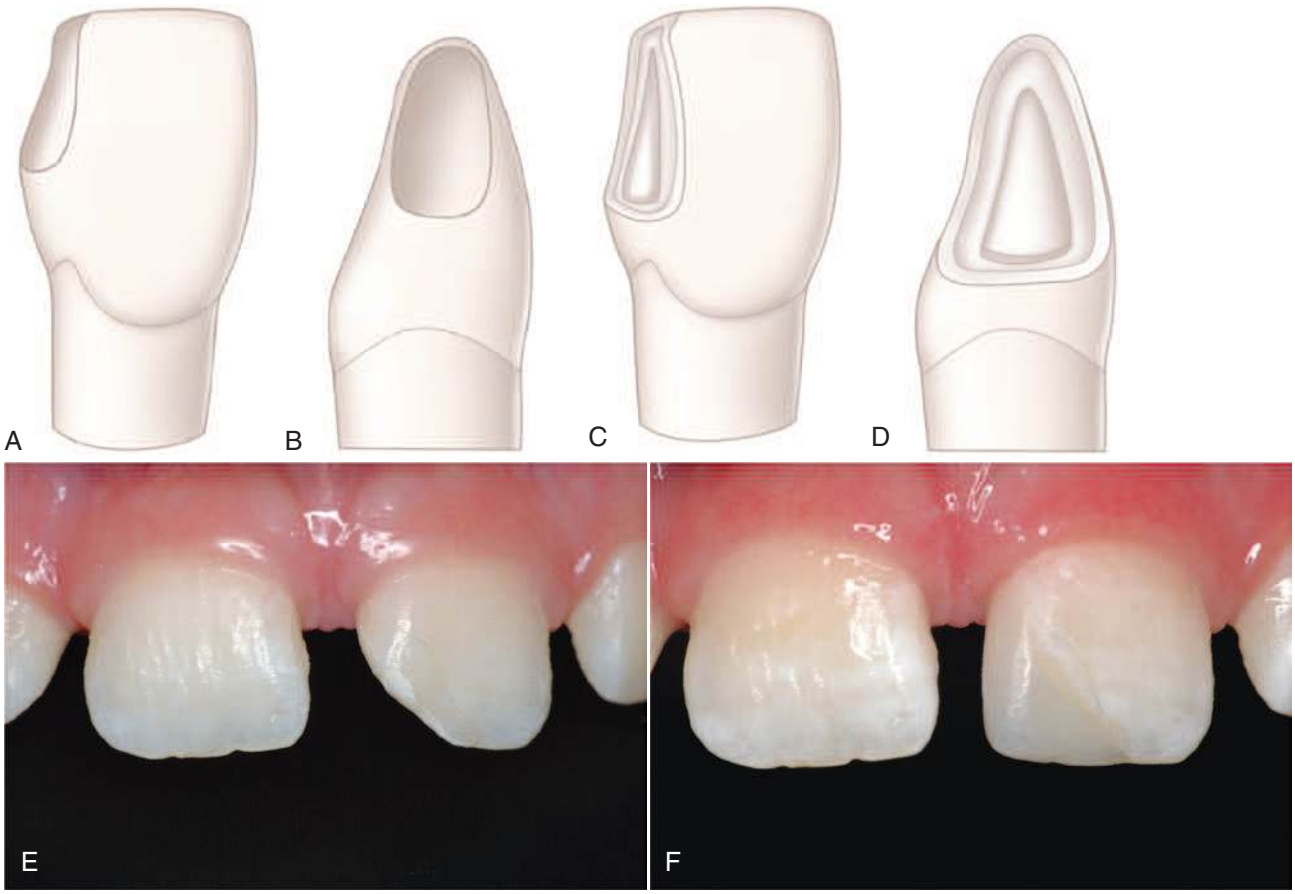
For large Class IV lesions or fractures, a preoperative impression may be taken to be used as a template for developing the restoration contours. This technique is described later in the chapter.

Tooth Preparation

Similar to the Class III preparation, the tooth preparation for a Class IV direct composite restoration involves (1) creating access to the defective structure (caries lesion, fracture, noncarious defect), (2) removing faulty structures (carious tissue removals, defective dentin and enamel, defective restoration, and base material), and (3) creating the convenience form for the restoration (Fig. 8.42A and B). The tooth preparation for large incisoproximal areas requires more attention to the retention form than that for a small Class IV defect (see Fig. 8.42C and D). If a large amount of tooth structure is missing and the restoration is in a high occlusal stress area, groove retention form may be indicated even when the preparation periphery is entirely in enamel. Also, enamel bevels can be increased in width to provide greater surface area for etching, resulting in a stronger bond between the composite and the tooth, and potentially a better esthetic result. Fig. 8.42E and F show a direct composite restoration of a Class IV defect.

The treatment of teeth with minor coronal fractures requires minimal preparation. If the fracture is confined to enamel, adequate retention usually can be attained by simply beveling the sharp cavosurface margins in the fractured area with a flame-shaped diamond instrument followed by bonding (Fig. 8.43). Regardless of its size, the extensions of the Class IV direct composite preparation are ultimately dictated by the extension of the caries lesion, fracture, or failed restoration being replaced. The outline form is prepared to include weakened, friable enamel.

Preparation of a maxillary right central incisor with a large Class IV defect and fractured mesioincisal corner, which necessitates a Class IV restoration, is illustrated in Fig. 8.44. The outline form is prepared using a round carbide bur or diamond instrument of



• **Fig. 8.42** Preparation designs for Class IV (A and B), and larger preparation designs for Class IV (C and D). E and F, Direct composite restoration of a Class IV defect, before (E) and after (F).

appropriate size at high speed with air-water coolant. All weakened enamel is removed, and the initial axial wall depth is established. As with the Class III tooth preparation, the final tooth preparation steps for a Class IV tooth preparation are, when indicated, (1) selective carious tissue removal (if present), (2) pulp protection (if needed), (3) bevel placement on accessible enamel margins, and (4) final procedures of cleaning and inspecting. The operator bevels the cavosurface margin of all accessible enamel margins of the preparation. The bevel is prepared at a 45-degree angle to the external tooth surface with a flame-shaped or round diamond instrument (see Fig. 8.44A). The width of the bevel should be 0.5 to 2 mm, depending on the amount of tooth structure missing and the retention perceived necessary. The use of a scalloped, nonlinear bevel sometimes helps in masking the restoration margin.

Although retention for most Class IV direct composite restorations is provided primarily by bonding of the composite to enamel and dentin, when large incisoproximal areas are being restored, additional mechanical retention may be obtained by groove-shaped or other forms of undercuts, dovetail extensions, or a combination of these. If retention undercuts are deemed necessary, a gingival retention groove is prepared using a No. ¼ round bur. It is prepared 0.2 mm inside the DEJ at a depth of 0.25 mm (half the diameter of the No. ¼ round bur). This groove should extend the length of the gingival floor and slightly up the facioaxial and linguoaxial line angles (see Fig. 8.44B). No retentive undercut is usually needed at the incisal area, where mostly enamel exists. Fig. 8.44C illustrates the completed large Class IV tooth preparation.

Restorative Technique

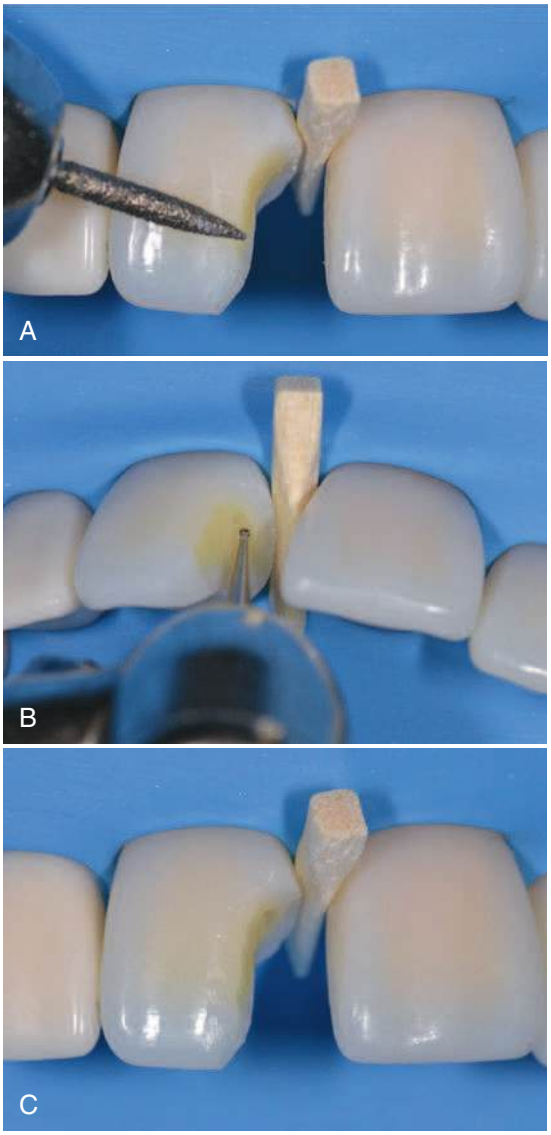
Matrix Application

Most Class IV composite restorations require a matrix to confine the restorative material and to assist in the development of the appropriate axial tooth contours, except for very small incisal edge enamel fractures, which can be restored using a free-hand technique. The Mylar matrix, described previously, also can be used for most Class IV preparations, although matrix flexibility makes control difficult. This difficulty may result in an overcontoured or undercontoured restoration, open contact, or both. Also, composite material extrudes incisally, but this excess can be easily removed when contouring and finishing.

Creasing (folding) the matrix at the position of the lingual line angle helps reduce the potential undercontouring (rounding) of that area of the restoration. The matrix is positioned and wedged as described for the Class III composite technique. Gingival overhangs and open contacts are common with any matrix technique that does not employ gingival wedging. A commercially available preformed plastic or celluloid crown form is usually too thick and is not recommended as a matrix. Alternatively, a custom lingual matrix may be used for large Class IV preparations.[57] Fig. 8.45A illustrates a large defective distofacial Class IV that needs to be replaced. The shade should be selected before isolating the area and removing the restorative material (see Fig. 8.45B). Fig. 8.45C shows a lingual view of the defective restoration after a rubber dam has been placed. Before the existing restoration is removed, the lingual matrix is prepared with either a polyvinyl siloxane



• **Fig. 8.43** Class IV tooth preparation and restoration. A, Extraoral view, minor traumatic fracture. B, Intraoral view. C, Fractured enamel is roughened with a flame-shaped diamond instrument. D, The conservative preparation is etched, while adjacent teeth are protected with Mylar strip. E and F, Contouring and polishing the composite. G, Intraoral view of the completed restoration. H, Extraoral view.



• **Fig. 8.44** Class IV tooth preparation. A, Beveling cavosurface margin of a large Class IV preparation. B, Gingival retention groove. C, Completed Class IV tooth preparation.

impression putty or a fast-set silicone matrix material. The operator records the lingual and incisal contours of the existing restoration by using a small amount of the silicone material, thus creating a guide or index with which the new restoration will be formed. The lingual matrix also can be fabricated from a quickly inserted mockup restoration or waxed study model for more complex cases when tooth structure is missing preoperatively. When obtained, the lingual matrix is set aside until it is time to insert the composite. Fig. 8.45D-E illustrate how the lingual matrix fits with the tooth preparation.

Placement of the Adhesive

Steps for placing the adhesive are the same as previously described for the Class III composite restoration. Also, the same considerations presented previously are appropriate for whether or not the matrix is placed before or after adhesive placement. When the lingual matrix technique is used, adhesive placement should occur before seating the matrix.

Insertion and Light Curing of Composite

After application of the adhesive, the operator inserts the composite with a hand instrument or a syringe as described earlier for Class III restorations. The composite is inserted in increments less than 2 mm thick. It is usually helpful to develop the lingual surface of the restoration first, then its body, and finally the facial surface. This approach facilitates the development of adequate anatomy with less potential for resultant excess composite material. This anatomic incremental layering also facilitates the development of adequate shade characterization, as dentin and enamel composite shades can be applied according to the structure they are replacing.

When using a properly contoured Mylar matrix, care must be taken when closing the matrix not to pull with excessive force because the soft material is extruded incisally and results in an undercontoured restoration. When this happens, composite should be added to restore proper contour and contact.

When a custom lingual matrix is used, the operator positions the lingual matrix and inserts the initial composite increment against the matrix in the lingual part of the preparation (see Fig. 8.45E). This initial increment, when polymerized, generally establishes the lingual, proximal, and incisal contours of the final restoration (see Fig. 8.45F). The operator continues to place and light cure additional increments until the desired form is obtained (see Fig. 8.45G). During these additions, color modifiers or tints may be incorporated to enhance the natural blue opalescence between tooth lobes or other subtle esthetic effects (see Fig. 8.45H). Because light-cured composites possess the advantage of an extended working time, the material can be manipulated and shaped to a considerable degree before light curing. Incremental insertion of light-cured composites also enables the clinician to layer composites with different optical properties (more opaque, darker in color, or both, to mimic dentin; and more translucent, lighter in color, or both, to mimic enamel), which can result in natural-looking composite restorations. After polymerization, the lingual matrix is removed. To ensure optimal polymerization, the operator light cures the restoration from facial and lingual directions. The final restoration is illustrated in Fig. 8.45I and J. This technique is particularly useful when the lingual contour of an existing composite restoration is to be duplicated in a new composite restoration, such as the case described. The technique facilitates the development of proper lingual, incisal, and proximal forms, reducing the need for contouring of the restoration.

Finishing and Polishing of the Composite

Finishing and polishing the Class IV composite are similar to the technique described for a Class III composite but usually more difficult. The primary differences are the involvement of the incisal angle and incisal edge of the tooth and an extended facial surface in large Class IV restorations. Finishing and polishing these sections of the restoration require similar procedural steps but close assessment of the incisal edge length and thickness, as well as of the facial macroanatomy and microanatomy of the tooth being restored. The functional demands of anterior guidance will have a direct influence on the final position and contours of the lingual surface and incisal edge. Careful adjustment and refinement will prevent excessive occlusal loading of the restoration and limit the potential for premature clinical failure. The facial, lingual, and proximal areas are finished and polished as described previously.



• **Fig. 8.45** Custom lingual matrix. A, Facial preoperative view. B, Preoperative shade determination. C, Lingual preoperative view after placement of the rubber dam. D, The old composite material is removed, and a conservative enamel bevel is placed. E, The lingual matrix obtained before tooth preparation is positioned and guides the application of the first lingual composite layer. F, The lingual composite layer determines the future contours of the restoration; note the intrinsic material translucency. Custom lingual matrix. G, The dentin buildup can be made directly against the lingual enamel; the clinician can visualize the whole tooth shape and place dentin with appropriate thickness and relation to the incisal edge. H, Color modifier or tint blue material is applied between the dentin lobes and slightly below the incisal edge to simulate the blue natural opalescence. I, View of the completed restoration (with second enamel layer placed on the buccal surface), after finishing. J, Facial postoperative view. (Courtesy of Dr. Didier Dietschi.)

Clinical Technique for Class V Direct Composite Restorations

Initial Clinical Procedures

The same initial procedure considerations presented for Class III restorations apply for Class V restorations, except for occlusal evaluation, which is not required for Class V restorations unless there are concerns about occlusal factors as an etiology for noncarious cervical lesions. During shade selection, it should be noted that the tooth is typically darker and more opaque in the cervical third. Isolation may be achieved by a rubber dam and No. 212 retainer or with a cotton roll and retraction cord.

Tooth Preparation

Class V tooth preparations, by definition, are located in the gingival one third of the facial and lingual tooth surfaces. Because of esthetic considerations, composites most frequently are used for the restoration of cervical caries lesions in anterior and premolar teeth. Numerous factors, including esthetics, nature of lesion and substrate characteristics, caries activity, caries risk, access to the lesion, and moisture control must be taken into consideration in material selection.

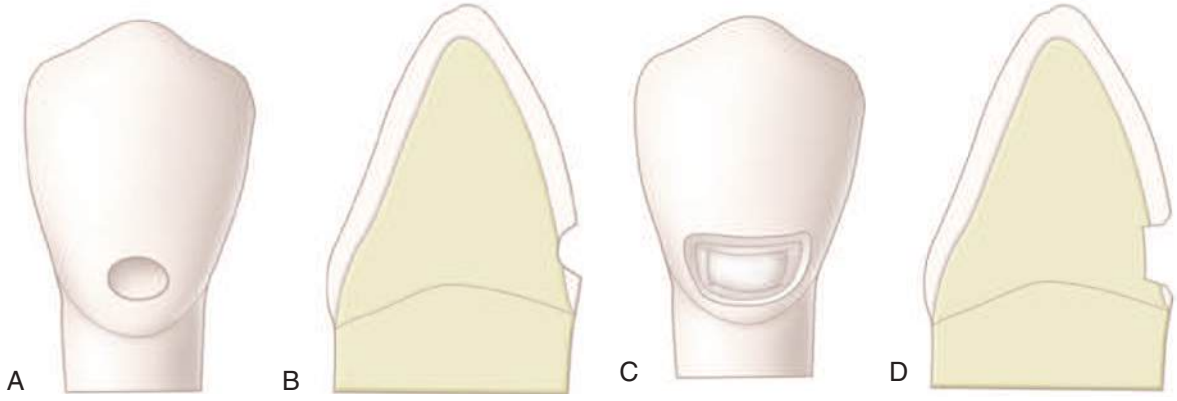
Because many Class V restorations involve root surfaces, at least on their cervical margins, careful consideration should be given to the choice of restorative material. The use of materials other than composite (i.e., that do not require adhesive placement) should be considered when factors that may compromise the performance of composite restorations are present. These factors include decreased salivary function, decreased patient motivation or ability for home care, increased difficulty in adequately isolating the operating area, and increased difficulty in performing the operative procedure because of the patient's physical or medical problems. Despite these concerns, the use of composite as a restorative material for cervical caries lesions predominates in areas of esthetic concern.

Similar to the Class III and IV preparations, tooth preparation for a Class V direct composite restoration involves (1) creating access to the defect (caries lesion, noncarious defect), (2) removing the defect (caries lesion, defective dentin and enamel, defective restoration and base material), and (3) creating the convenience form for the restoration. The tooth preparation for large Class V lesions or defects may require more attention to retention form than for conservative Class V defects, especially when little enamel is available for bonding. Areas of hypermineralized (sclerotic) dentin also may require special attention, as these respond differently to bonding than areas with normal dentin.[58-60] Enamel bevels are typically used on the occlusal/incisal margins of the preparation, while at the cervical margin, bevels are usually not recommended because of the absence of enamel in this area. Class V tooth preparations will vary slightly, depending on the type and extension of the defect being restored.

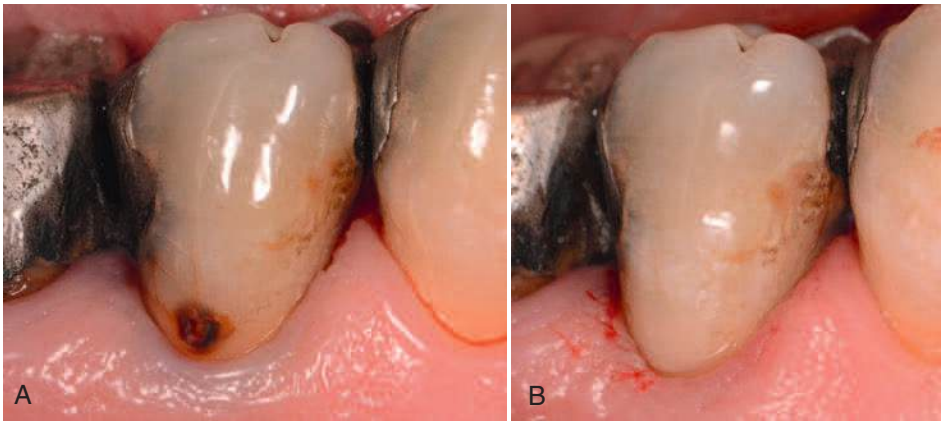
The objective of the *Class V tooth preparation for small or moderate lesions or defects that do not extend onto the root surface* is to restore the lesion or defect as conservatively as possible. No effort is made to prepare the walls as butt joints, and usually no secondary retentive features are incorporated. The lesion or defect is conservatively prepared resulting in a form that may have a divergent wall configuration and an axial surface that usually is not uniform in depth (Fig. 8.46A and B). Small or moderate Class V tooth preparations are ideal for small enamel defects or small primary caries lesions (Fig. 8.47A). These include decalcified and hypoplastic areas located in the cervical third of the teeth, which may be restored for esthetic reasons. The typical outline form for a Class V lesion in enamel is shown in Fig. 8.48.

After the usual preliminary procedures, the initial tooth preparation is accomplished with a round diamond or carbide bur, eliminating the entire enamel lesion or defect. The preparation is extended into dentin only when the defect warrants such extension. No effort is made to prepare 90-degree cavosurface margins. If carious tissue remains in dentin, it is selectively removed with a round bur or spoon excavator. Usually, this preparation technique will result in a slightly beveled enamel margin. If deemed necessary, the enamel margin can be further beveled. The completed restoration is shown in Fig. 8.47B.

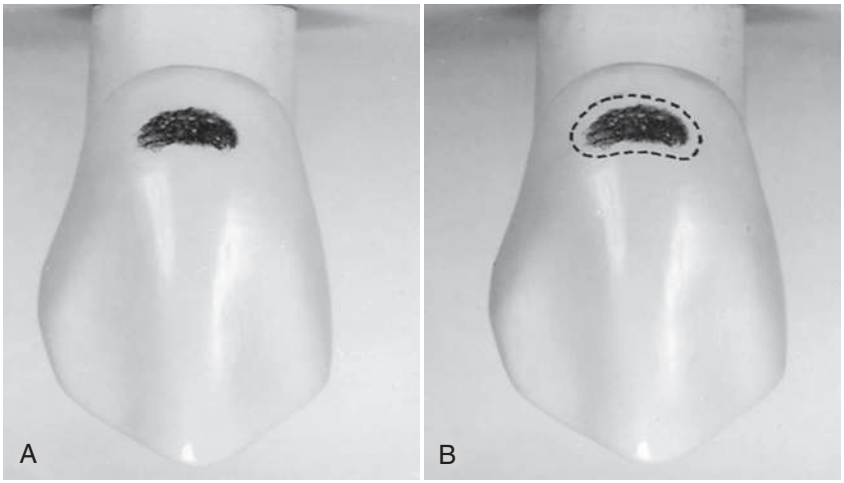
An example of a small cervical caries lesion was presented in Fig. 8.49A, which illustrates a path of a decalcified enamel lesion (in enamel only) having a broken, rough surface that extends mesially or distally from the cavitated lesion (or failing existing restoration). After preparation of the cavitated lesion (or failing restoration), the margins of the preparation are extended to include these areas of decalcification by using a round diamond or bur to prepare the cavosurface margin in the form of a chamfer, extended in the enamel only to a depth that removes the defect. A completed preparation of this type was illustrated in Fig. 8.49B.



• **Fig. 8.46** Preparation designs for Class V (A and B) initial composite restorations (primary caries) and larger preparation designs for Class V (C and D) restorations.



• **Fig. 8.47** Class V tooth preparation. A, Small cavitated Class V lesion. B, Restored with composite resin. (Courtesy Vilhelm G. Ólafsson.)



• **Fig. 8.48** A, Class V caries. B, Typical outline form.

Small or moderate Class V tooth preparations also are used to restore noncarious cervical lesions (NCCLs). As noted elsewhere in this textbook, NCCLs have a multifactorial etiology in which contributing factors include intrinsic and/or extrinsic erosion, abrasion, abfraction, and/or biocorrosion.[61] If the causative factors are not eliminated, these lesions tend to progress and enlarge with time. When they are detected (Fig. 8.50A), the operator first must decide, with input from the patient, whether the area needs to be restored. This decision is based on the following considerations.

Caries

If active caries is present, caries management should be initiated. For the incipient lesion, treatment may consist of application of a topical fluoride or resin-based materials including adhesives and resin infiltration materials. Inactive lesions may be monitored and kept unrestored indefinitely, but sometimes these can be esthetically unsatisfactory. (Most NCCLs are not carious.)

Gingival Health

If the defect is determined to be contributing to the development of gingival inflammation and/or gingival recession (i.e., plaque retention, mechanical trauma due to poor tooth contours), the defect should be restored.

Esthetics

If the defect is in an esthetically critical position, the patient may elect to have the area restored with a tooth-colored restoration.

Sensitivity

If the defect is very sensitive, application of an adhesive or desensitizing agent may reduce or eliminate the sensitivity, at least temporarily. Continuing sensitivity may require restoration of the area.

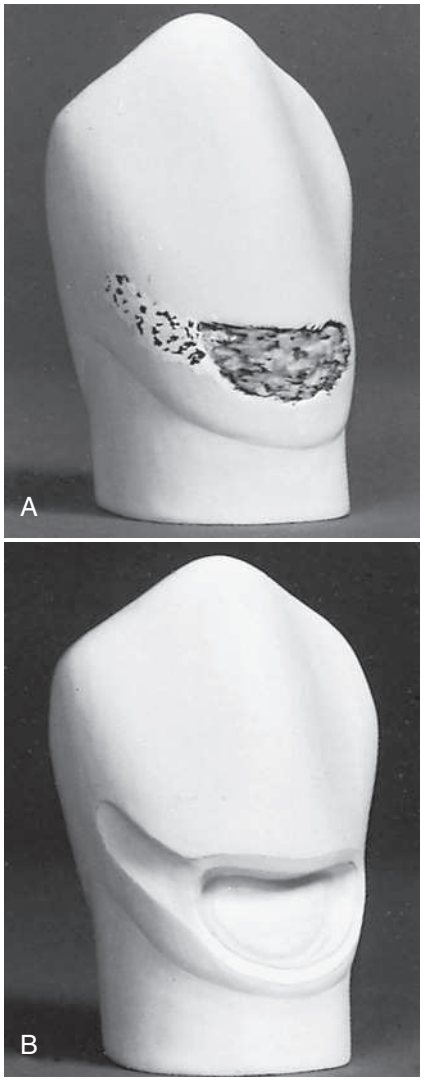
Pulp Protection

If the defect is very large and deep pulpally, its restoration may be indicated to avoid further defect development that may cause a pulpal exposure.

Tooth Strength

If the defect is very large or deep, the strength of the tooth at the cervical area may be compromised. Placement of a bonded restoration reduces the potential for further progression of the defect and may restore some of the lost strength.

Periodontal therapy with gingival grafts also may be considered as a treatment option for these exposed root surfaces being used solo or combined with the restorative therapy.



• **Fig. 8.49** A, Decalcified area extending mesially from cavitated Class V lesion. B, Completed Class V preparation with conservative mesial extension.

The tooth preparation for a *Class V NCCL* usually requires only surface debridement of any exposed dentin and roughening/beveling all enamel margins with a diamond instrument[22] (see Fig. 8.50B). Often, because of the inherent form of an abraded or eroded lesion, further preparation of root surface cavosurface margins is not needed. The completed preparation with etched enamel is shown in Fig. 8.50C.

In *Class V tooth preparation for large lesions or defects that extend onto the root surface* (Fig. 8.51A), the gingival aspect of the preparation form is similar to that described in Chapter 10 for a Class V amalgam restoration. The features of the preparation include a 90-degree cavosurface margin. Groove retention form usually is not necessary but may be used if retention of the restoration is a concern. The enamel margins are prepared using the same design described for small or moderate Class V tooth preparations, that is, with a conservative enamel bevel. This preparation design is indicated for the replacement of an existing, defective Class V restoration that initially used a conventional preparation or for a large, new caries lesion. The large Class V preparation initially exhibits 90-degree cavosurface margins (that subsequently can be beveled when in enamel) and an axial wall that may or may not be uniform in depth (see Fig. 8.47C and D). The axial depth into dentin is determined by the selective removal of the carious tissue. Many of these larger preparations are a combination of beveled enamel margins and 90-degree root surface cavosurface margins. A completed large Class V preparation extending onto the root surface is illustrated in Fig. 8.52.

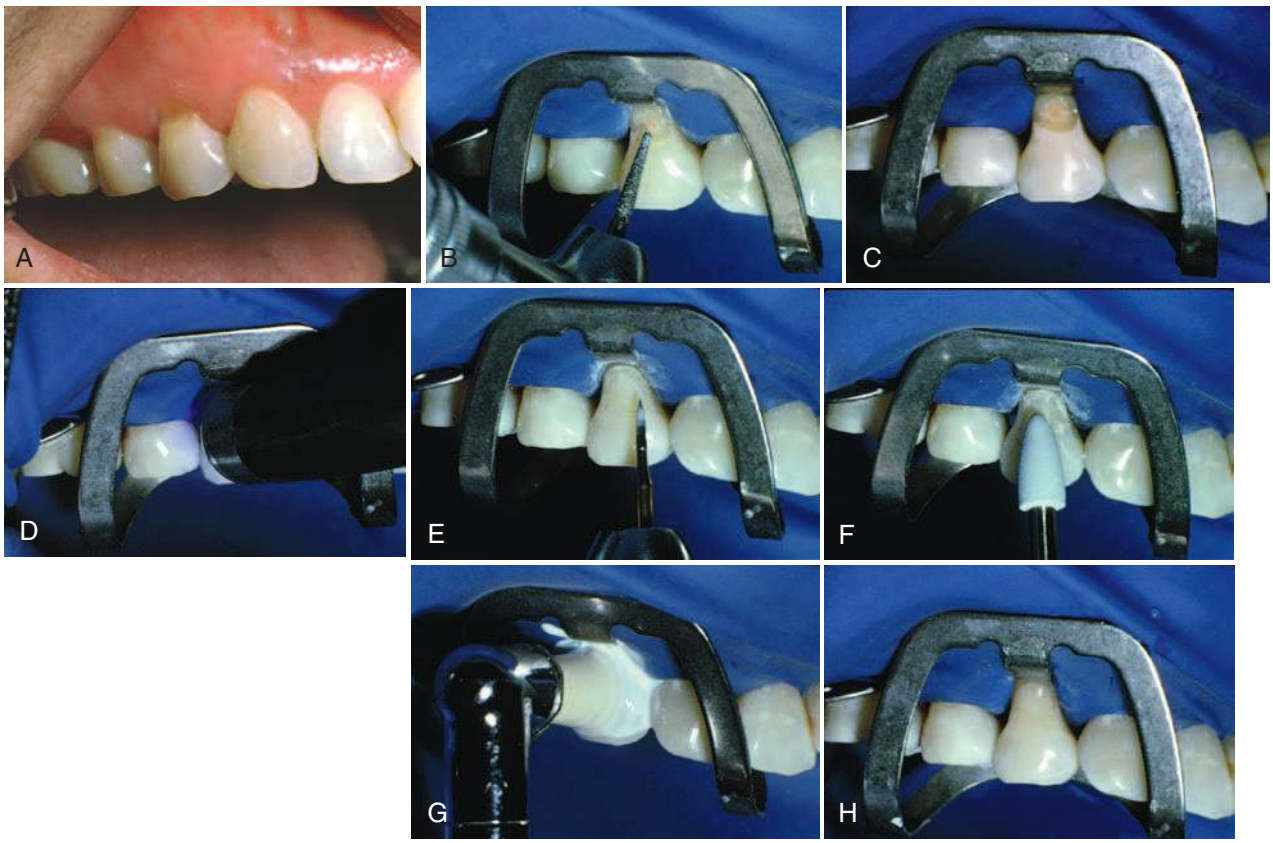
To initiate the preparation, a tapered fissure carbide bur (No. 271) or similarly shaped diamond is used at high speed with air-water spray. If interproximal or gingival access is limited, an appropriately sized round bur or diamond may be used. When a tapered fissure bur or diamond is used, the handpiece is maneuvered to maintain the bur's long axis perpendicular to the external surface of the tooth during preparation of the outline form, which should result in 90-degree cavosurface margins. At this initial tooth preparation stage, the extensions in every direction are to sound tooth structure. Deeper areas of the caries lesion or old restoration may still remain in the axial wall at this point in the procedure (see Fig. 8.51B). Carious tissue remaining on this initial axial wall is selectively removed during the final stage of tooth preparation (see Fig. 8.51C). Old restorative material remaining may or may not be removed according to the concepts stated previously. When the desired distal extension is obtained, the instrument is moved mesially, incisally (occlusally), and gingivally for indicated extensions, while maintaining proper depth and the instrument's long axis perpendicular to the external surface. The axial wall should follow the original contour of the facial or lingual surface, which is convex outward mesiodistally and sometimes occlusogingivally. The outline form extension of the mesial, distal, occlusal (incisal), and gingival walls is dictated by the extent of the caries lesion, defect, or old restorative material indicated for replacement (sometimes the new material abuts a still satisfactory old restoration).

All of the external preparation walls of a Class V tooth preparation are visible when viewed from a facial position (outwardly divergent walls). Final tooth preparation consists of the following steps: (1) removing the remaining carious dentin or old restorative material (if indicated) on the axial wall; (2) applying a base, only if necessary, as discussed previously in this chapter; and (3) beveling the enamel margins and adding groove retention, if indicated. The bevel on the enamel margin is accomplished with a flame-shaped or round diamond instrument, resulting in an angle approximately 45 degrees to the external tooth surface, and prepared to a width of at least 0.5 mm depending on the preparation size and esthetic considerations. A completed Class V preparation is shown in Fig. 8.53.

Occasionally, a tooth surface that normally is smooth has a pit in the enamel (Fig. 8.54A). Most unusual pit faults in enamel are restored best with the preparation design described for small or moderate Class V defects. For such a preparation for an unusual pit fault, the outline form (extensions and depth) is dictated by the extent of the fault or caries lesion. Faults existing entirely in enamel are prepared with an appropriately sized round diamond instrument by merely eliminating the defect leaving a flared enamel margin (see Fig. 8.54B). Adequate retention is obtained by bonding. When the defect includes carious dentin, the tissue is selectively removed as indicated.

Restorative Technique

In most cases no matrix is needed for Class V restorations because the contour can be controlled as the composite restorative material is being inserted. However, clear precontoured Class V matrix materials are available if the operator prefers a matrix-assisted technique.



• **Fig. 8.50** Class V tooth preparation for abrasion and erosion lesions. A, Preoperative notched lesion. B, Beveling the enamel margin. C, Completed preparation with etched enamel. Restoration of a noncarious cervical lesion. D, After application of bonding adhesive, composite resin is inserted incrementally and the restorative material is light cured. Finishing and polishing. E, Flame-shaped finishing bur removing excess and contouring. F and G, Rubber polishing point (F) and aluminum oxide polishing paste (G) used for final polishing. H, Completed restoration.

Placement of the Adhesive

The techniques for acid etching of the involved tooth structure and placement of the adhesive are the same as previously described in this chapter.

Insertion and Light Curing of the Composite

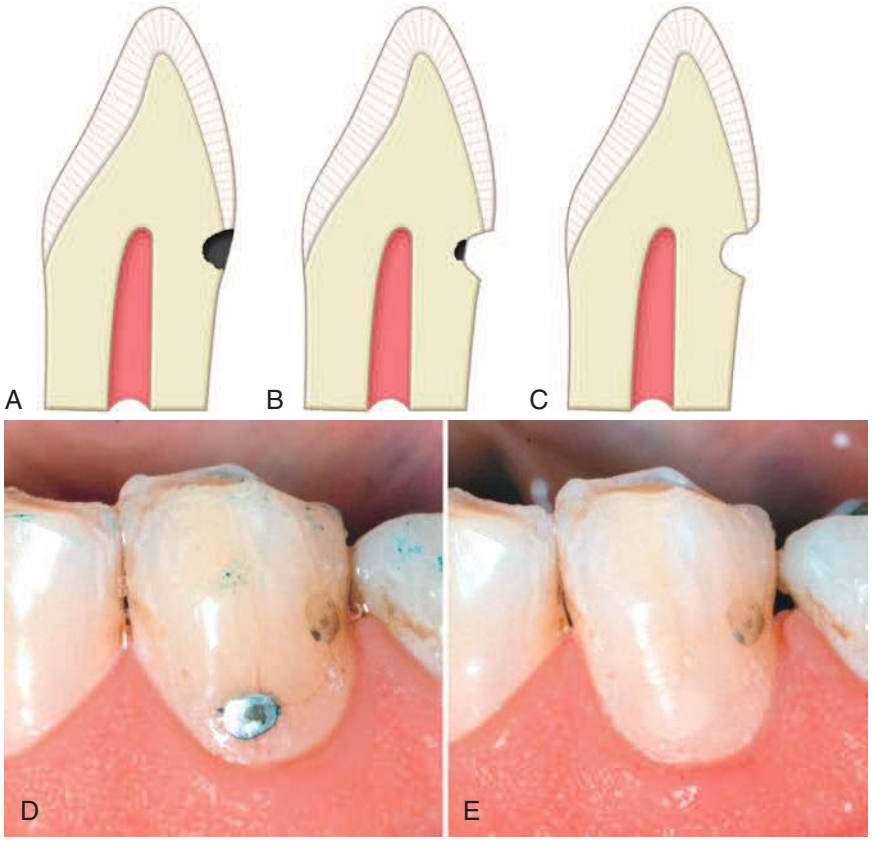
The composite can be inserted with a hand instrument or syringe. Composites and RMGIs are recommended for Class V restorations. A light-cured material is recommended for most Class V preparations because of the extended working time and control of contour before polymerization. Careful placement allows for minimal need of rotary finishing. These features are particularly valuable when restoring large preparations or preparations with margins located on cementum because rotary instrumentation can easily damage the contiguous tooth structure and/or compromise the marginal integrity of the restoration.

After bonding procedures (according to manufacturer's instructions), it is recommended to insert the composite incrementally with a hand instrument or syringe. The number and position of the increments depend on the size and depth of the preparation. For large and deep preparations, an incremental technique is recommended. Deep preparations with retentive undercuts are usually filled with at least two axially placed increments with minimal contact on the occlusal/incisal and gingival walls. The operator should avoid placing increments that connect to both occlusal/incisal walls and gingival walls, or mesial and distal walls, simultaneously, so as to minimize polymerization shrinkage stresses. Regardless of the technique used, before light curing the last increment that defines the contour of the restoration, the material is shaped as close to the final contour as possible. An explorer or blade of a composite instrument, or a flat No. 2 sable brush, is useful in removing excess material from the cervical margin and obtaining the final contour. As noted earlier, clear plastic matrices are available for Class V composite restorations and can be used if the operator prefers a matrix-assisted approach rather than a free-hand technique. The light source is applied for polymerization (see Fig. 8.50D). The restoration should require very little finishing.

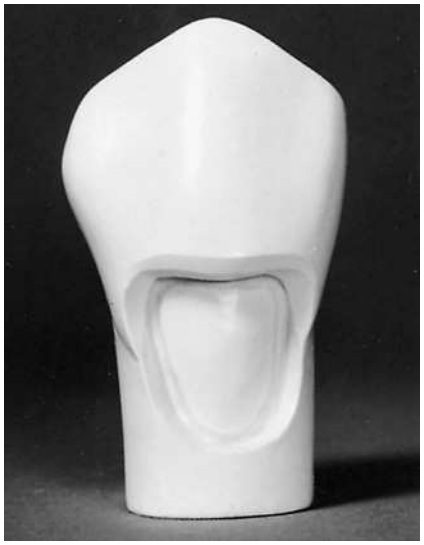
Finishing and Polishing of the Composite

When excess composite is present, a flame-shaped carbide finishing bur or diamond is recommended for removing excess composite on the facial surface (see Fig. 8.50E). Medium speed with light intermittent brush strokes and an air coolant are used for finishing the restoration to the correct contours. Final polishing is achieved with a rubber polishing point (see Fig. 8.50F) or cup, diamond-impregnated polisher, and sometimes a polishing paste (see Fig. 8.50G and H).

For some locations, abrasive discs, as stated earlier, mounted on an appropriate mandrel in an angled handpiece at low speed can be used (see Fig. 8.41A). The degree of abrasiveness depends



• **Fig. 8.51** Class V tooth preparation. A, Lesion extending onto root surface. B, Initial tooth preparation with 90-degree cavosurface margins and axial wall depth of 0.75 mm. C, Remaining infected dentin excavated, and incisal enamel margin beveled. D and E, Direct composite restoration of three Class V lesions with composite resin, before (D) and after (E). (D and E, Courtesy Vilhelm G. Ólafsson.)



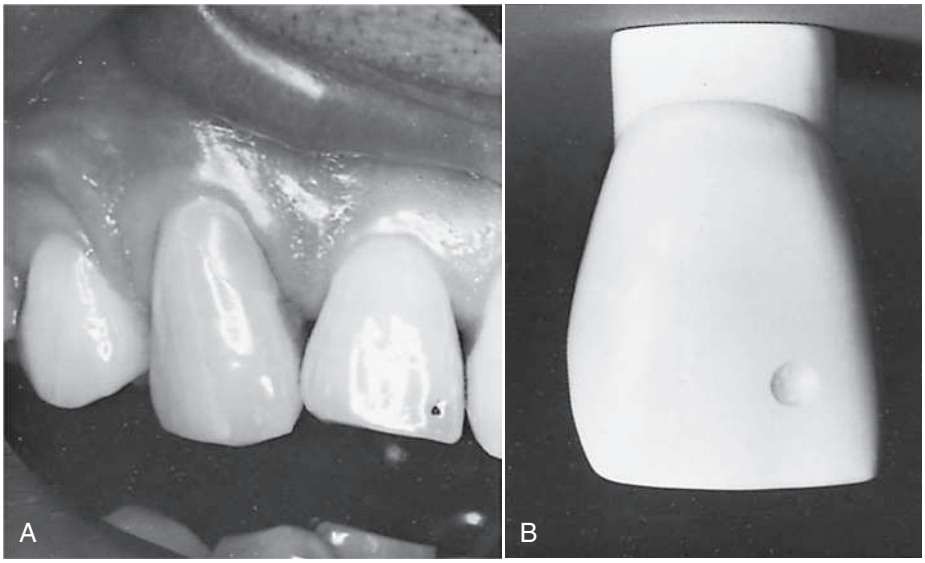
• **Fig. 8.52** Completed Class V tooth preparation extending onto the root; the incisal margin is beveled; the root portion has a retention groove for increased retention. Retention grooves on the incisal aspect are rarely required and are shown here for illustration purposes only.



• **Fig. 8.53** Completed large Class V preparation. A retention groove on the incisal aspect is rarely required and is shown here for illustration purposes only.

on the amount of excess to be removed. The external enamel surface should act as a guide for a proper contour, preventing the development of a flat surface contour.

Rotary instruments should be used carefully in gingival locations (especially on margins adjacent to the root surface) to prevent inadvertent and undesirable damage to tooth structure and periodontal tissues. Finely pointed rotary instruments (finishing burs or diamonds) are difficult to use to remove gingival margin excess. Because of the convexity that typically exists in this area, a more rounded rotary instrument (a fine diamond) may remove the excess



• **Fig. 8.54** A, Faulty pit on the facial surface of the maxillary incisor. B, Tooth preparation for an enamel pit defect.

with less potential to damage the unprepared root surface. Likewise, abrasive discs used on the root surface may cause ditching of the cementum, if not used correctly. Fig. 8.51D and E show restoration of a Class V lesion with composite resin.

Clinical Technique for Class VI Direct Composite Restorations

One of the most conservative indications for a directly placed posterior composite is a small faulty developmental pit located on a cusp tip. Fig. 8.55A is an example of a Class VI defect on the facial cusp tip of a maxillary premolar. The typical Class VI tooth preparation should be as small in diameter and as shallow in depth as possible. The faulty pit is entered with an appropriate round bur or diamond oriented perpendicular to the surface and extended pulpally to eliminate the lesion (see Fig. 8.55B). Visual examination and probing with an explorer often reveal that the fault is limited to enamel because the enamel in this area is quite thick. If a faulty restoration or extensive caries lesion is present on the cusp tip, a round bur of appropriate size is used to remove the faulty restoration or any remaining carious tissue. Stains that appear through the translucent enamel should be removed; otherwise they may be seen after the composite restoration is completed. Some undermined, but not friable, enamel may be left and bonded to the composite in areas that are not associated with a functional occlusal contact.

The conservative preparation is restored with composite following adhesion protocols, incremental placement and light curing of the composite material, and finishing and polishing. The final Class VI restoration should, as with any other type of restoration, restore the tooth to its ideal anatomy relative to tooth contours, occlusion, esthetics, and surface characteristics (Fig. 8.55C).

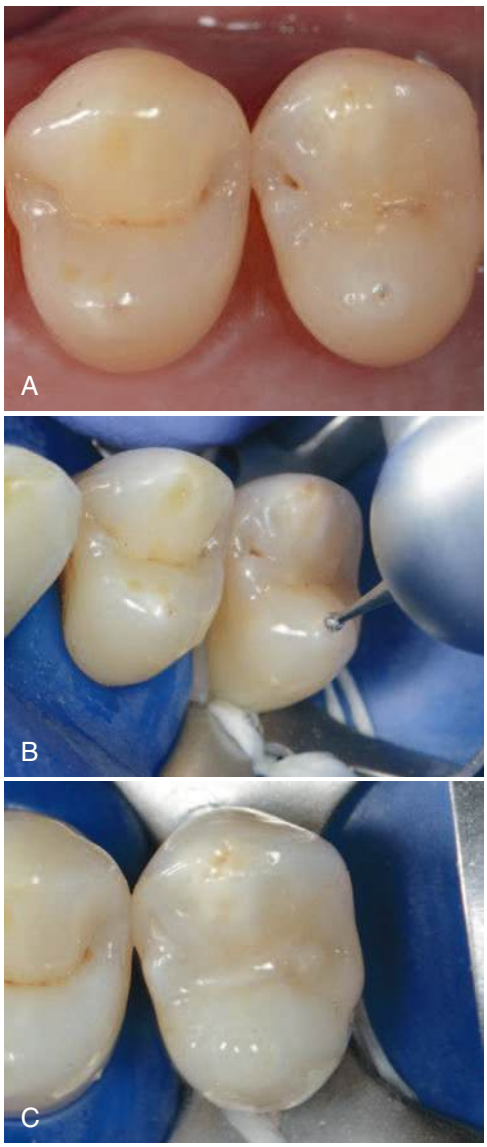
Clinical Technique for Glass Ionomer Restorations

Glass ionomers possess the favorable quality of releasing fluoride when exposed to the oral environment.[62,63] These materials also have been shown to "recharge" with fluoride when exposed to fluoride from various sources.[64] These properties theoretically may indicate the use of glass ionomer restorations in patients with high caries risk especially when esthetic demands are minimal. (See Chapter 13 for types of glass ionomers.)

Because of their limited strength and wear resistance, glass ionomers are indicated generally for the restoration of low-stress areas (not for typical Class I, II, or IV restorations), where caries activity potential is of significant concern. In addition to being indicated for root-surface caries lesions in Class V locations, slotlike preparations in Class II or III cervical locations (not involving the proximal contact) may be restored with glass ionomers, if access permits.

The restoration of caries lesions on the roots of patients with active caries is the primary indication for the use of a glass ionomer as a restorative material. Cervical defects of idiopathic erosion or abrasion origin (or any combination) also may be indications for restoration with glass ionomers, if esthetic demands are not critical. The tooth preparations for either of these clinical indications are the same as previously described for composite restorations (see Figs. 8.50, 8.51, and 8.52), except bevels are rarely used. Gingival recession and interproximal gingival inflammation also increase the risk of developing caries on proximal root surfaces. Gingival recession sometimes provides access to this type of caries lesion from the facial or lingual direction, allowing a slot preparation to be used. The same slot preparation design used for amalgam is used for glass ionomers. (The reader is referred to the sections on slot preparations in Chapter 10 and to Fig. 8.56 for specific details.) With the exception of the matrix used (if needed), slot preparations for Class II and III restorations are restored in a single increment and in a similar manner to a Class V preparation.

Most conventional glass ionomer systems require mild dentin conditioning to remove the smear layer, effecting improved adhesion of the glass ionomer to dentin. To condition dentin, a mild acid, such as 10% polyacrylic acid, is applied to the preparation according to manufacturer's instructions, followed by rinsing and removal of excess water, leaving dentin slightly moist. Some modified glass ionomer materials may have a substantial resin component



• **Fig. 8.55** Class VI tooth preparation for composite restoration. A, Class VI preparation on the lingual cusp tip of the maxillary premolar. B, Entry with small round bur or diamond. C, Final composite restoration and sealant placed in the occlusal groove. (Courtesy Vilhelm G. Ólafsson.)

and require a special primer to facilitate bonding. Each system should be used strictly according to the manufacturer's specific instructions.

Encapsulated glass ionomers, for triturator mixing, or paste–paste materials are greatly preferred to powder and liquid materials. Such systems optimize and simplify the mixing procedure. Glass ionomer material should be placed into the preparation in slight excess and quickly shaped with a composite instrument. Clear plastic cervical matrices also are available for providing contour to the restoration. If a conventional glass ionomer is used, a thin coat of light-cured, resin-based coating is placed on the surface immediately after placement to limit dehydration and resultant disintegration ("cracking") of the restoration during the initial setting phase. Newer glass ionomers may be more resistant to dehydration; however, careful attention to manufacturer's instructions for restoration placement is paramount.

Most current restorative GICs and RMGIs available may be finished and polished immediately after light or complete chemical/dual curing. When the material has set, the matrix, if used, is removed and the gross excess is shaved away with a No. 12 surgical blade. Finishing should be accomplished as much as possible with hand instruments, while striving to preserve the smooth surface that occurs on setting. If rotary instrumentation is needed, care must be taken not to dehydrate the surface of the restoration. Also, flexible abrasive discs used with a lubricant can be effective. A fine-grit aluminum oxide polishing paste applied with a prophy cup is used to impart a smooth surface. Fig. 8.57 shows a conventional glass ionomer restoration with a modified matrix band. The Tofflemire matrix band was adapted around the tooth and an access hole was created using a No. 2 round bur. The conventional glass ionomer material was then injected into the preparation through the access hole until the material filled into the preparation space and excess material extruded through the access. Sufficient time was allowed for the material to set before finishing the surface with a No. 12 blade.

Repairing Composite Restorations

If a patient presents with a composite restoration that has a localized defect, a repair usually can be made.[65,66] Easily accessible areas may be roughened with a diamond; the area is etched; an appropriate adhesive is applied; and the composite is inserted, finished, and polished. If the defect is not easily accessible, a tooth preparation must be created that exposes the defective area, and a matrix may be necessary; the adhesive and the composite are then placed.

If a void is detected immediately after insertion of a composite restoration, but before contouring is initiated, more composite may be added directly to the void area. These materials bond because the void area has an oxygen-inhibited surface layer that permits composite additions. If any contouring has occurred, however, the oxygen-inhibited layer may have been removed or contaminated, and the area must be "refreshed" by a brief etch/rinse cycle and reapplication of the adhesive prior to the addition of more composite.

Common Problems: Causes and Potential Solutions

This section lists the causes of common problems associated with some composite restorations and potential solutions to those problems.

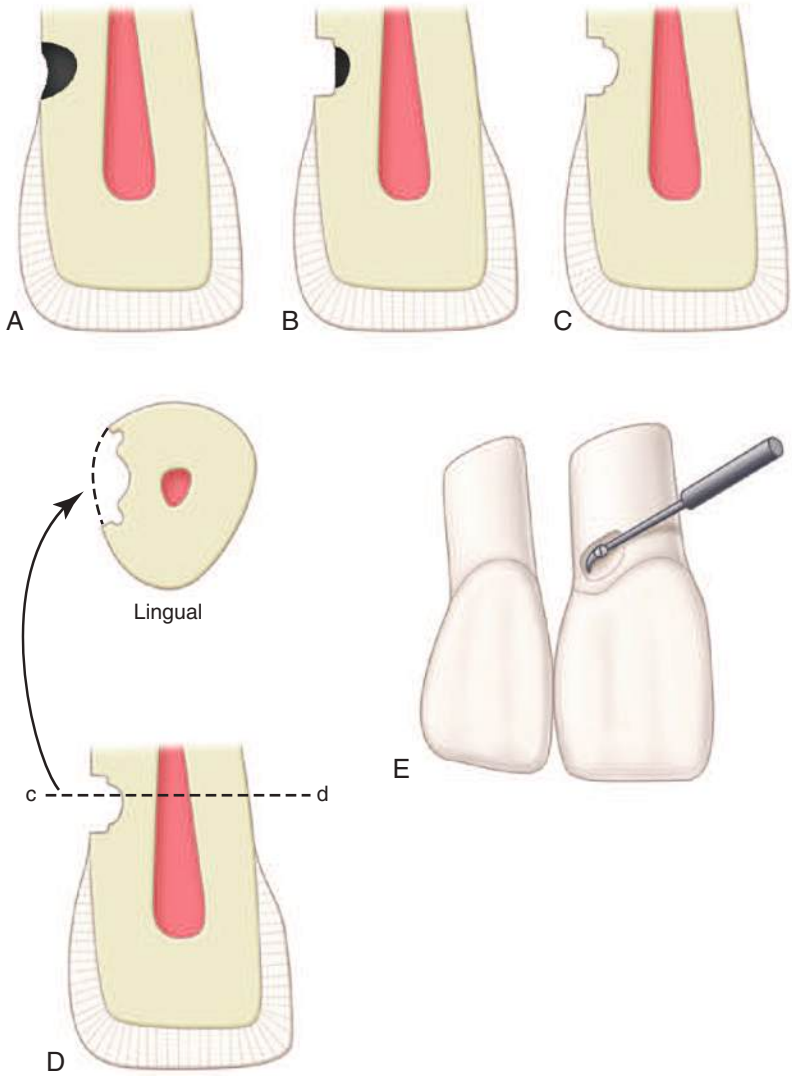
White Line Adjacent to the Enamel Margin

A white line adjacent to the enamel margin of a direct composite restoration is usually caused by a microseparation (space) between the composite and the tooth or a microfracture within the marginal enamel, either of which is typically caused by:

- Inadequate etching and bonding of the affected area
- High-intensity, fast light curing resulting in excessive polymerization stresses
- Traumatic finishing techniques

Potential solutions are as follows:

- Use atraumatic finishing techniques (e.g., light intermittent pressure)
- Use proper polymerization techniques (see Chapter 6)



• **Fig. 8.56** Class III tooth preparation for a lesion entirely on the root surface. A, Mesiodistal longitudinal section illustrating a caries lesion. B, Initial tooth preparation. C, Tooth preparation with infected caries dentin removed. D, Retention grooves shown in longitudinal section. Transverse section through plane *cd* illustrates the contour of the axial wall and the direction of the facial and lingual walls. E, Preparing the retention form to complete the tooth preparation.

Some of these small defects may not be clinically relevant and can be monitored over time for signs of marginal deterioration. However, often they need to be managed, as follows:

- Seal the gap with adhesive
- Conservatively remove the defect and re-restore

Voids

Causes of voids include the following:

- Spaces left between increments during insertion (lamination defects)
- Adherence of composite to hand instruments during placement in the preparation, which is related to the previous item

Potential solutions are as follows:

- Use a more careful restorative technique
- Repair marginal voids by preparing the area and re-restoring

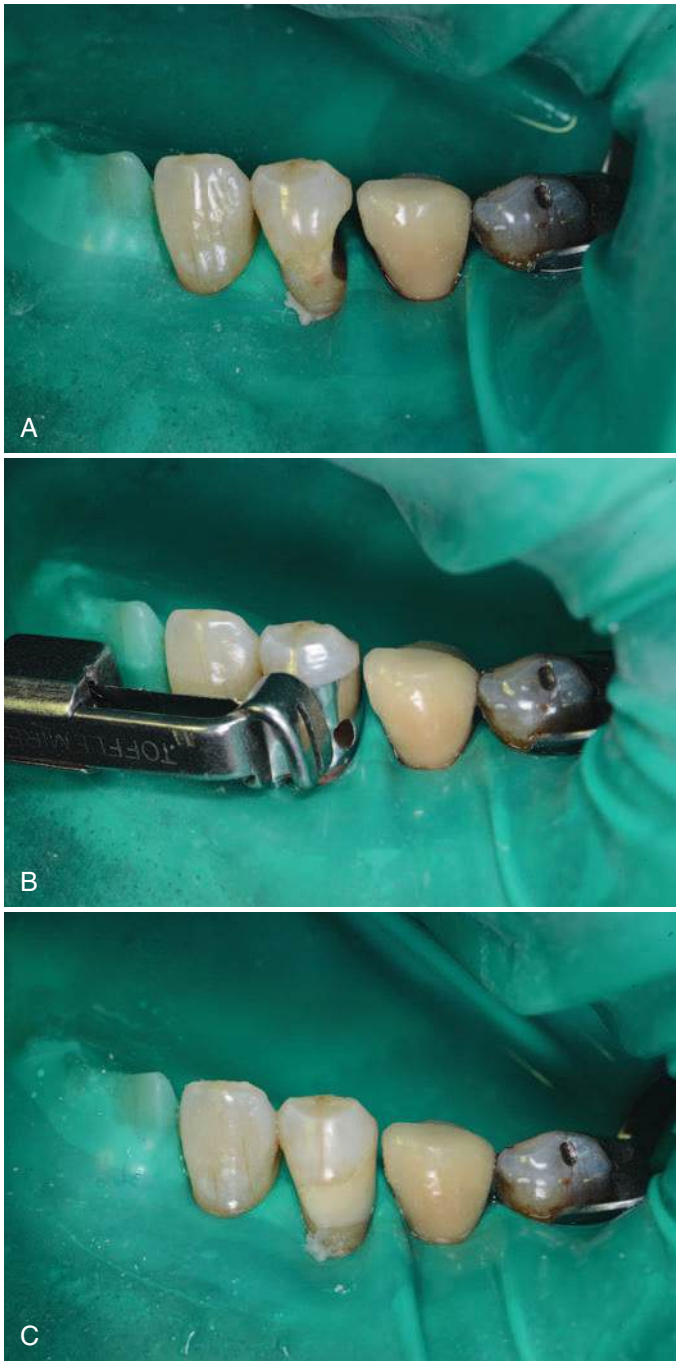
Weak or Missing Proximal Contacts (Class II, III, and IV)

Causes of weak and missing proximal contacts are as follows:

- Inadequately contoured matrix band
- Inadequate wedging, preoperatively and during the composite insertion
- Matrix band movement during composite insertion or matrix band not in direct contact with the adjacent proximal surface
- A circumferential matrix being used when restoring only one contact
- Adherence of composite to hand instruments during insertion and resultant pulling away from matrix contact area
- Matrix band too thick

Potential solutions include the following:

- Contour the matrix material properly
- Have the matrix in contact with the adjacent tooth



• **Fig. 8.57** Conventional glass ionomer restoration with a modified matrix band is shown before (A), matrix band adapted with access hole prepared into the band (B), and immediately postop (C). (Courtesy Vilhelm G. Ólafsson.)

- Use adequate preoperative and insertion wedging technique
- Use a matrix system that places the matrix only around the proximal surface to be restored
- Use a hand instrument to hold the matrix against the adjacent tooth while curing incremental placements of composite
- Be careful with insertion technique

Inaccurate Shade

Causes of an incorrect shade matching include the following:
- Inappropriate operatory lighting while selecting the shade
- Selection of shade after the tooth has dehydrated
- Shade tab not matching the actual composite shade
- Wrong shade utilized

Potential solutions are as follows:
- Use natural light when selecting shade, if possible
- Select the shade before isolating the tooth
- Place some of the selected shade on the tooth and cure preoperatively to verify shade selection
- Do not shine the operator light (overhead or attached to operator loupes) directly on the area during shade selection
- Understand the typical zones of different shades for natural teeth

Contouring and Finishing Problems

Causes of contouring and/or finishing problems are as follows:
- Inadequate anatomic tooth form (overcontour or undercontour of the restoration)
- Improper selection of finishing instrument and improper composite placement
- Iatrogenic damage to adjacent unprepared tooth structure

Potential solutions include the following:
- Have a proper matrix with appropriate axial and line angle contours
- Create embrasures to match the adjacent tooth embrasure form
- Remember the outline form of the preparation
- Use a properly shaped finishing instrument for the area being contoured
- Be careful with the use of rotary instruments to avoid adversely affecting the structure of the adjacent tooth or teeth
- Do not use rotary instruments that leave roughened surfaces

Postoperative Sensitivity

Postoperative sensitivity can result from the following:
- Aggressive tooth preparation (inadequate use of coolant systems, inefficient cutting instruments)
- Incorrect use of adhesive systems
- Not using a liner/base when indicated
- Formation of microgaps secondary to excessive polymerization shrinkage stress, particularly in situations of high C-factor
- Aggressive finishing of the restoration

Potential solutions include the following:
- Use careful tooth preparation technique
- Use liners/bases properly
- Use adhesive systems properly
- Use a desensitizer solution after dentin acid etching
- Insert and polymerize the composite material properly
- Finish the composite restoration properly

Summary

Direct composite restorations are increasingly used because of the benefits accrued from adhesive bonding to tooth structure, esthetic qualities, and almost universal clinical use. When accomplished properly, a composite restoration is able to provide excellent service for many years. Composite is the material of choice for most Class III and IV restorations and most esthetically demanding Class V restorations. When used in Class I and II restorations, however, composites are more difficult and sensitive to the operator's technique than are amalgam restorations. To achieve optimal clinical results, the operating site must be free from contamination, and the bonding technique and composite handling must be meticulous in every way.

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